

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	4
Nozzle Schedule:	6
Bill of Material:	8
MDMT Summary:	9
Warnings and Errors:	11
Input Echo:	12
XY Coordinate Calculations:	24
Internal Pressure Calculations:	25
Element: Left Head	26
Element: Shell-1	28
Element: Shell-2	30
Element: Shell-3	32
Element: Shell-4	34
Element: Right Head	36
Element: Boot Head	38
Shop Pressure Test Results:	41
External Pressure Calculations:	43
Element and Detail Weights:	44
Nozzle Flange MAWP:	47
Wind Load Calculation:	48
Earthquake Load Calculation:	51
Center of Gravity Calculation:	53
Saddle Calcs: Operating Case:	54
Saddle Calcs: Test Case:	68
Nozzle Calcs.: Nozzle_A_24"	82
Nozzle Calcs.: Nozzle_B_24"	89
Nozzle Calcs.: BOOT	96
Nozzle Calcs.: L2_2"	100
Nozzle Calcs.: L4_2"	105
Nozzle Calcs.: L6_2"	110
Nozzle Calcs.: L8_2"	115
Nozzle Calcs.: K9 2"	120
Nozzle Calcs.: K10 2"	125
Nozzle Calcs.: L1_2"	130
Nozzle Calcs.: L3_2"	135
Nozzle Calcs.: L5_2"	140
Nozzle Calcs.: L7_2"	145
Nozzle Calcs.: MH-1_24"	150
Nozzle Calcs.: Nozzle_C_10"	155



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\146-D - Rev 01.pvdb

Date of Analysis : Jul 30, 2026 11:50am

PV Elite 28 U2, April 2026



Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 3048.000 mm ID
 Vessel Design Length, Tangent to Tangent 990.60 cm
 Specified Datum Line Distance 0.00 cm
 Internal Design Temperature 82 °C
 Internal Design Pressure 184.204 kgf/cm²
 External Design Temperature 82 °C
 Maximum Allowable Working Pressure 185 kgf/cm²
 Shop Test Pressure 249.226 kgf/cm²
 Required Minimum Design Metal Temperature -33.0 °C
 Warmest Computed Minimum Design Metal Temperature 16.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	+IT
Nozzle	SA-765 I	...	...	K03046	
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Left Head	184.387	0.00	185.000	3.1750	N/A	No
Shell-1	184.387	0.00	185.000	3.1750	N/A	No
Shell-2	184.387	0.00	185.000	3.1750	N/A	No
Shell-3	184.387	0.00	185.000	3.1750	N/A	No
Shell-4	184.387	0.00	185.000	3.1750	N/A	No
Right Head	184.387	0.00	185.000	3.1750	N/A	No
Boot Head	184.342	0.00	187.000	3.1750	No	No

Liquid Level: 68.60 cm Dens.: 657.651 kg/m³ Sp. Gr.: 0.658

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Sphere	0.0	0.0	102.0	94.0	93.6	...	1.00	1.00
Cylinder	300.0	300.0	185.0	185.0	184.0	...	1.00	1.00
Cylinder	600.0	300.0	185.0	185.0	184.0	...	1.00	1.00
Cylinder	900.0	300.0	185.0	185.0	184.0	...	1.00	1.00



Cylinder	990.6	90.6	185.0	185.0	184.0	...	1.00	1.00
Sphere	990.6	0.0	102.0	94.0	93.6	...	1.00	1.00
Ellipse	100.0	0.0	36.0	30.0	29.7	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Nozzle_A_24"	WRC-107/537	0.996	n/a	Passed
Nozzle_B_24"	WRC-107/537	0.996	n/a	Passed
Nozzle_C_10"	WRC-107/537	0.988	n/a	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	280.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	2134.000	mm
Wear Pad Width	305.000	mm
Wear Pad Thickness	26.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1981.000	mm
Baseplate Length	2780.000	mm
Baseplate Thickness	30.000	mm
Baseplate Width	305.000	mm
Number of Ribs (including outside ribs)	6	
Rib Thickness	20.000	mm
Web Thickness	20.000	mm
Height of Center Web	370.000	mm
Number of Bolts in Baseplate	8	

Baseplate Sketch

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	143984.33	kgf
Maximum Transverse Saddle Shear Load	10404.78	kgf
Maximum Longitudinal Saddle Shear Load	20809.56	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	159272.14	kgf
Maximum Transverse Saddle Shear Load	711.68	kgf
Maximum Longitudinal Saddle Shear Load	755.03	kgf

Weights:

Fabricated - Bare w/o removable internals	207499.2	kg
Shop Test - Fabricated + water (full)	296960.7	kg
Shipping - Fab. + rem. intls.+ shipping app.	216216.9	kg
Erected - Fab. + rem. intls.+ insul. (etc)	217236.9	kg
Empty - Fab. + intls. + details + wghts.	217236.9	kg
Operating - Empty + operating liquid (no ca)	226548.7	kg

Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
Nozzle_A_24"	185	...	...	OK	OK	Passed
Nozzle_B_24"	185	...	...	OK	OK	Passed
BOOT	185	...	...	OK	OK	Passed
L2_2"	192	...	...	OK	OK	Passed
L4_2"	192	...	...	OK	OK	Passed
L6_2"	192	...	...	OK	OK	Passed
L8_2"	192	...	...	OK	OK	Passed
K9_2"	185	...	...	OK	OK	Passed
K10_2"	185	...	...	OK	OK	Passed
L1_2"	185	...	...	OK	OK	Passed
L3_2"	185	...	...	OK	OK	Passed
L5_2"	185	...	...	OK	OK	Passed
L7_2"	185	...	...	OK	OK	Passed
MH-1_24"	185	...	...	...	OK	Passed
Nozzle_C_10"	189	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 185 Nozzle : MH-124"
 Minimum MAWP Shells/Flanges : 185 Element : Right Head
 Minimum MAPnc Shells/Flanges : 204 Element : Shell-4

Computed Vessel M.A.W.P. : 185 kgf/cm²

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

- Error: Nozzle Nozzle_A_24", was too close to a gross structural discontinuity.
- Error: Nozzle Nozzle_B_24", was too close to a gross structural discontinuity.
- Error: Nozzle Nozzle_C_10", was too close to a gross structural discontinuity.

Please see ASME VIII-2 Section 4.5.13.1 for more information.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
10	Nozzle_A_24"	0.000	180.000	263.975
10	Nozzle_B_24"	0.000	0.000	263.975
20	BOOT	1000.000	270.000	257.175
20	L2_2"	0.000	270.000	28.575
20	L4_2"	0.000	250.000	28.575
20	L6_2"	0.000	220.000	28.575
20	L8_2"	0.000	200.000	28.575
20	K9_2"	1500.000	90.000	25.692
20	K10_2"	2000.000	90.000	25.692
30	L1_2"	3500.005	0.000	18.575
30	L3_2"	4000.005	0.000	18.575
30	L5_2"	4500.005	0.000	18.575
30	L7_2"	5000.005	0.000	18.575
60	MH-1_24"	0.000	0.000	263.975
70	Nozzle_C_10"	0.000	0.000	110.175

The nozzle spacing is computed by the following:

$$= \sqrt{l_l^2 + l_c^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

PV Elite 28 U2 Licensee: SL User
FileName : 146-D - Rev 01
Nozzle Summary:

Page 5 of 161

Step: 37 11:50am Jul 30, 2026



HEXAGON

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
L2_2"	2.000 in	Actual	LWN	2.787	10.000	...	...	355.45	1500
L4_2"	2.000 in	Actual	LWN	2.787	10.000	...	...	355.45	1500
L6_2"	2.000 in	Actual	LWN	2.787	10.000	...	...	355.45	1500
L8_2"	2.000 in	Actual	LWN	2.787	10.000	...	...	355.45	1500
K9 2"	2.000 in	160	None	2.375	8.738	...	...	489.16	...
K10 2"	2.000 in	160	None	2.375	8.738	...	...	489.16	...
L1_2"	2.000 in	Actual	WNF	2.000	10.000	...	...	488.65	1500
L3_2"	2.000 in	Actual	WNF	2.000	10.000	...	...	488.65	1500
L5_2"	2.000 in	Actual	WNF	2.000	10.000	...	...	488.65	1500
L7_2"	2.000 in	Actual	WNF	2.000	10.000	...	...	488.65	1500
Nozzle_C_10"	10.000 in	Actual	None	10.000	20.000	...	...	367.82	...
BOOT	20.000 in	Actual	None	23.465	44.000	...	...	864.19	...
Nozzle_A_24"	24.000 in	Actual	None	24.000	44.000	...	...	1024.32	...
Nozzle_B_24"	24.000 in	Actual	None	24.000	44.000	...	...	1024.32	...
MH-1_24"	24.000 in	Actual	WNF	24.000	44.000	...	...	663.21	1500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
L2_2"	SA-765 I	184.000	10.000	...	...	...
L4_2"	SA-765 I	184.000	10.000	...	...	...
L6_2"	SA-765 I	184.000	10.000	...	...	...
L8_2"	SA-765 I	184.000	10.000	...	...	...
K9 2"	SA-765 I	184.000	10.000	...	...	...
K10 2"	SA-765 I	184.000	10.000	...	...	...
L1_2"	SA-765 I	184.000	10.000	...	...	...
L3_2"	SA-765 I	184.000	10.000	...	...	...
L5_2"	SA-765 I	184.000	10.000	...	...	...
L7_2"	SA-765 I	184.000	10.000	...	...	...
Nozzle_C_10	SA-765 I	30.000	10.000	...	...	...
BOOT	SA-765 I	184.000	10.000	...	...	...
Nozzle_A_24	SA-765 I	94.000	10.000	...	...	...
Nozzle_B_24	SA-765 I	94.000	10.000	...	...	...
MH-1_24"	SA-765 I	94.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
L2_2"	20.000	270.0	300.00	0.00	BOOT
L4_2"	20.000	250.0	300.00	0.00	BOOT
L6_2"	20.000	220.0	300.00	0.00	BOOT
L8_2"	20.000	200.0	300.00	0.00	BOOT
K9 2"	150.000	90.0	300.00	0.00	Shell-1
K10 2"	200.000	90.0	300.00	0.00	Shell-1
L1_2"	350.000	0.0	300.00	0.00	Shell-2



L3_2"	400.000	0.0	300.00	0.00	Shell-2
L5_2"	450.000	0.0	300.00	0.00	Shell-2
L7_2"	500.001	0.0	300.00	0.00	Shell-2
Nozzle_C_10"	...	0.0	300.00	0.00	Boot Head
BOOT	100.000	270.0	610.00	0.00	Shell-1
Nozzle_A_24"	...	180.0	500.00	0.00	Left Head
Nozzle_B_24"	...	0.0	500.00	0.00	Left Head
MH-1_24"	...	0.0	500.00	0.00	Right Head



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 102.0mm THK X 3137.0mm ID X 0.0cm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 36.0mm THK X 508.0mm ID X 10.0cm	SA-516 70
1	PLATFORM: 1000mm X 3416mm WIDE	
1	LADDER: 3500.0mm	...
2	INSULATION: 1568mm X 60mm THK	
2	SADDLE: 280.0mm X 120 DEG	SA-516 70
3	INSULATION: 3000mm X 60mm THK	
1	INSULATION: 906mm X 60mm THK	
1	INSULATION: 227mm X 109mm THK	
1	CLASS 1500 GR 1.1, 24.0" BLIND FLANGE(S)	SA-350 LF2
9	CYLINDRICAL SEGMENT 185.0mm THK X 3048.0mm X 300.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 185.0mm THK X 431.6mm X 300.0cm	SA-516 70
2	CYLINDRICAL SEGMENT 185.0mm THK X 6527.5mm X 300.0cm	SA-516 70
3	CYLINDRICAL SEGMENT 185.0mm THK X 3048.0mm X 90.6cm	SA-516 70
1	CYLINDRICAL SEGMENT 185.0mm THK X 6527.5mm X 90.6cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
Left Head	[9]	!	-33	Yes			
Shell-1	[8]	!	-33	Yes			
Shell-2	[8]	!	-33	Yes			
Shell-3	[8]	!	-33	Yes			
Shell-4	[8]	!	-33	Yes			
Right Head	[9]	!	-33	Yes			
Boot Head	[10]	B	17	No	0.914	30.000	1.00
Boot Head	[7]	B	21	No	0.856	36.000	1.00
NozzleA24"	[1]	B	20	Yes	0.917	94.000	1.00
NozzleB24"	[1]	B	20	Yes	0.917	94.000	1.00
BOOT	[1]			Yes			
L22"	[1]	B	7	Yes	0.882	44.000	1.00
Nozzle Flg	[5]	B	-46	Yes			
L42"	[1]	B	7	Yes	0.882	44.000	1.00
Nozzle Flg	[5]	B	-46	Yes			
L62"	[1]	B	7	Yes	0.882	44.000	1.00
Nozzle Flg	[5]	B	-46	Yes			
L82"	[1]	B	7	Yes	0.882	44.000	1.00
Nozzle Flg	[5]	B	-46	Yes			
K9 2"	[1]	B	-104	Yes	0.041	90.000	1.00
K10 2"	[1]	B	-104	Yes	0.041	90.000	1.00
L12"	[1]	B	-20	Yes	0.381	10.000	1.00
Nozzle Flg	[5]	B	-46	Yes			
L32"	[1]	B	-20	Yes	0.381	10.000	1.00
Nozzle Flg	[5]	B	-46	Yes			
L52"	[1]	B	-20	Yes	0.381	10.000	1.00
Nozzle Flg	[5]	B	-46	Yes			
L72"	[1]	B	-20	Yes	0.381	10.000	1.00
Nozzle Flg	[5]	B	-46	Yes			
MH-124"	[1]	B	20	Yes	0.917	94.000	1.00
Nozzle Flg	[5]	B	-46	Yes			
NozzleC10"	[1]	B	17	No	0.910	30.000	1.00
Nozl. Bolting	[22]		-48	No	1.000	64.000	1.00

Warmest MDMT: 21 16

Required Minimum Design Metal Temperature -33.0 °C

Warmest Computed Minimum Design Metal Temperature 16.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld





Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in. ⁴	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in. ³	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	184.2	kgf/cm ²
Design Internal Temperature	82.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-33.0	°C
Type of Construction	Welded	
Special Service	None	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW



Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 97.2	km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	0	cm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 cm
Distance Upwind of Crest	Lh	0 cm
Distance from Crest to the Vessel	x	0 cm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	cm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration	[Ss] 0.147
Max. Eff. Ground Acceleration	[S] 0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	Left Head
Distance "FROM" to "TO"	0 cm
Inside Diameter	3137 mm
Element Thickness	94 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	102 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	184.34 kgf/cm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	82 °C



Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Allowable Stress, Ambient	1784.4	kgf/cm ²
Allowable Stress, Operating	1647.5	kgf/cm ²
Allowable Stress, Hydrotest	2518.6	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2473.7	kgf/cm ²
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Platform	
Detail ID	PLATFORM	
Dist. from "FROM" Node / Offset dist	0	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	111680	cm ²
Platform Weight	5888.7	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	3416	mm
Platform Height	1000	mm
Platform Clearance or End Offset	95	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	350	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	9908	mm
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	68.6	cm
Liquid Density	657.65	kg/m ³
Element From Node	10	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-156.85	cm
Height/Length of Insulation	156.85	cm
Thickness of Insulation	60	mm
Density	0.00032	kg/cm ³
Element From Node	10	
Detail Type	Nozzle	
Detail ID	NozzleA24"	
Dist. from "FROM" Node / Offset dist	800	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-765 I	
Element From Node	10	
Detail Type	Nozzle	
Detail ID	NozzleB24"	
Dist. from "FROM" Node / Offset dist	800	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	



Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	None
Nozzle Matl	SA-765 I
Element From Node	10
Detail Type	Weight
Detail ID	cut-back
Dist. from "FROM" Node / Offset dist	0 cm
Miscellaneous Weight	1600 kgf
Offset from Element Centerline	0 mm

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	300 cm
Inside Diameter	3048 mm
Element Thickness	185 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	185 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	184.34 kgf/cm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	198.1 cm
Width of Saddle	280 mm
Height of Saddle at Bottom	2134 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	305 mm
Thickness of Wear Plate	26 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 cm

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	68.6 cm
Liquid Density	657.65 kg/m ³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	300 cm
Thickness of Insulation	60 mm
Density	0.00032 kg/cm ³



Element From Node 20
 Detail Type Nozzle
 Detail ID BOOT
 Dist. from "FROM" Node / Offset dist 100 cm
 Nozzle Diameter 20 in.
 Nozzle Schedule None
 Nozzle Class None
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 kgf
 Grade of Attached Flange None
 Nozzle Matl SA-765 I

Element From Node 20
 Detail Type Nozzle
 Detail ID L22"
 Dist. from "FROM" Node / Offset dist 20 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-765 I

Element From Node 20
 Detail Type Nozzle
 Detail ID L42"
 Dist. from "FROM" Node / Offset dist 20 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 250.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-765 I

Element From Node 20
 Detail Type Nozzle
 Detail ID L62"
 Dist. from "FROM" Node / Offset dist 20 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 220.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-765 I

Element From Node 20
 Detail Type Nozzle
 Detail ID L82"
 Dist. from "FROM" Node / Offset dist 20 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 200.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-765 I

Element From Node 20
 Detail Type Nozzle
 Detail ID K9 2"
 Dist. from "FROM" Node / Offset dist 150 cm
 Nozzle Diameter 2 in.



Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-765 I	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K10 2"	
Dist. from "FROM" Node / Offset dist	200	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-765 I	

Element From Node	20	
Detail Type	Weight	
Detail ID	6"pipe support	
Dist. from "FROM" Node / Offset dist	198.1	cm
Miscellaneous Weight	39.999	kgf
Offset from Element Centerline	0	mm

Element From Node	20	
Detail Type	Weight	
Detail ID	Anti-Swirl Baff	
Dist. from "FROM" Node / Offset dist	100	cm
Miscellaneous Weight	39.999	kgf
Offset from Element Centerline	1400	mm

Element From Node	20	
Detail Type	Weight	
Detail ID	Misc weight	
Dist. from "FROM" Node / Offset dist	150	cm
Miscellaneous Weight	9999.8	kgf
Offset from Element Centerline	0	mm

Element From Node	20	
Detail Type	Weight	
Detail ID	Trunnion1	
Dist. from "FROM" Node / Offset dist	100	cm
Miscellaneous Weight	1420	kgf
Offset from Element Centerline	1700	mm

Element From Node	20	
Detail Type	Weight	
Detail ID	Trunnion2	
Dist. from "FROM" Node / Offset dist	100	cm
Miscellaneous Weight	1420	kgf
Offset from Element Centerline	1700	mm

Element From Node	20	
Detail Type	Weight	
Detail ID	Fire proofing	
Dist. from "FROM" Node / Offset dist	198.1	cm
Miscellaneous Weight	210	kgf
Offset from Element Centerline	2000	mm

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Shell-2



Distance "FROM" to "TO"	300	cm
Inside Diameter	3048	mm
Element Thickness	185	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	185	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	184.34	kgf/cm ²
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	68.6	cm
Liquid Density	657.65	kg/m ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	300	cm
Thickness of Insulation	60	mm
Density	0.00032	kg/cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	L12"	
Dist. from "FROM" Node / Offset dist	50	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 I	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	L32"	
Dist. from "FROM" Node / Offset dist	100	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 I	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	L52"	
Dist. from "FROM" Node / Offset dist	150	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 I	



Element From Node	30	
Detail Type	Nozzle	
Detail ID	L72"	
Dist. from "FROM" Node / Offset dist	200	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 I	

Element From Node	30	
Detail Type	Weight	
Detail ID	demister-1	
Dist. from "FROM" Node / Offset dist	150	cm
Miscellaneous Weight	200	kgf
Offset from Element Centerline	1000	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	demister-2	
Dist. from "FROM" Node / Offset dist	150	cm
Miscellaneous Weight	99.998	kgf
Offset from Element Centerline	900	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	Pipe+Expander	
Dist. from "FROM" Node / Offset dist	150	cm
Miscellaneous Weight	4799.9	kgf
Offset from Element Centerline	0	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	Shroud+Spt Clea	
Dist. from "FROM" Node / Offset dist	150	cm
Miscellaneous Weight	699.99	kgf
Offset from Element Centerline	100	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	Demister Grid1	
Dist. from "FROM" Node / Offset dist	150	cm
Miscellaneous Weight	49.999	kgf
Offset from Element Centerline	900	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	Demister Grid2	
Dist. from "FROM" Node / Offset dist	150	cm
Miscellaneous Weight	49.999	kgf
Offset from Element Centerline	800	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	Distributor pla	
Dist. from "FROM" Node / Offset dist	250	cm
Miscellaneous Weight	299.99	kgf
Offset from Element Centerline	1200	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	Side Baffles	
Dist. from "FROM" Node / Offset dist	200	cm
Miscellaneous Weight	150	kgf
Offset from Element Centerline	0	mm



Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	300 cm
Inside Diameter	3048 mm
Element Thickness	185 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	185 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	184.34 kgf/cm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Right Saddle
Dist. from "FROM" Node / Offset dist	192.5 cm
Width of Saddle	280 mm
Height of Saddle at Bottom	2134 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	305 mm
Thickness of Wear Plate	26 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 cm

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	68.6 cm
Liquid Density	657.65 kg/m ³

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	300 cm
Thickness of Insulation	60 mm
Density	0.00032 kg/cm ³

Element From Node	40
Detail Type	Weight
Detail ID	6" pipe support
Dist. from "FROM" Node / Offset dist	192.5 cm
Miscellaneous Weight	39.999 kgf
Offset from Element Centerline	0 mm

Element From Node	40
Detail Type	Weight
Detail ID	Ins Cleat
Dist. from "FROM" Node / Offset dist	150 cm
Miscellaneous Weight	100 kgf
Offset from Element Centerline	0 mm

Element From Node	40
-------------------	----



Detail Type	Weight
Detail ID	Fire Proofing
Dist. from "FROM" Node / Offset dist	192.5 cm
Miscellaneous Weight	210 kgf
Offset from Element Centerline	2000 mm

Element From Node	40
Detail Type	Weight
Detail ID	Platform Cleat
Dist. from "FROM" Node / Offset dist	150 cm
Miscellaneous Weight	180 kgf
Offset from Element Centerline	0 mm

Element#	5/7
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell-4
Distance "FROM" to "TO"	90.6 cm
Inside Diameter	3048 mm
Element Thickness	185 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	185 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	184.34 kgf/cm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	68.6 cm
Liquid Density	657.7 kg/m ³

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	90.6 cm
Thickness of Insulation	60 mm
Density	0.00032 kg/cm ³

Element From Node	50
Detail Type	Weight
Detail ID	Trunnion3
Dist. from "FROM" Node / Offset dist	45.001 cm
Miscellaneous Weight	1420 kgf
Offset from Element Centerline	1700 mm

Element From Node	50
Detail Type	Weight
Detail ID	Trunnion4
Dist. from "FROM" Node / Offset dist	45 cm
Miscellaneous Weight	1420 kgf
Offset from Element Centerline	1700 mm

Element#	6/7
Element From Node	60
Element To Node	70



Element Type	Spherical
Description	Right Head
Distance "FROM" to "TO"	0 cm
Inside Diameter	3137 mm
Element Thickness	94 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	102 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	184.34 kgf/cm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	68.6 cm
Liquid Density	657.65 kg/m ³

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 60
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	156.85 cm
Thickness of Insulation	60 mm
Density	0.00032 kg/cm ³

Element From Node	60
Detail Type	Nozzle
Detail ID	MH-124"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-765 I

Element From Node	60
Detail Type	Weight
Detail ID	Manway Davit
Dist. from "FROM" Node / Offset dist	156 cm
Miscellaneous Weight	200 kgf
Offset from Element Centerline	0 mm

Element From Node	60
Detail Type	Weight
Detail ID	cut back
Dist. from "FROM" Node / Offset dist	0 cm
Miscellaneous Weight	1600 kgf
Offset from Element Centerline	0 mm

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	Boot Head
Distance "FROM" to "TO"	10 cm
Inside Diameter	508 mm
Element Thickness	30 mm



Internal Corrosion Allowance	3.175	mm
Nominal Thickness	36	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	184.34	kgf/cm ²
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	-12.7	cm
Height/Length of Insulation	22.7	cm
Thickness of Insulation	110	mm
Density	0.00032	kg/cm ³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	NozzleC10"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	10	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-765 I	



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Left Head		...	...	...	...
Shell-1		300	...	300	...
Shell-2		600	...	300	...
Shell-3		900	...	300	...
Shell-4		990.6	...	90.6	...
Right Head		990.6	...	...	...
Boot Head		100	-213.4	...	10

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Left Head		184.39	102	3.175	3137	1647.5	1
Shell-1		184.39	185	3.175	3048	1647.5	2
Shell-2		184.39	185	3.175	3048	1647.5	3
Shell-3		184.39	185	3.175	3048	1647.5	4
Shell-4		184.39	185	3.175	3048	1647.5	5
Right Head		184.39	102	3.175	3137	1647.5	6
Boot Head		184.34	36	3.175	508	1647.5	7

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Left Head		184.342	185.069	207.715	94	93.6336	1
Shell-1		184.342	185.279	204.44	185	184.029	2
Shell-2		184.342	185.279	204.44	185	184.029	3
Shell-3		184.342	185.279	204.44	185	184.029	4
Shell-4		184.342	185.279	204.44	185	184.029	5
Right Head		184.342	185.069	207.715	94	93.6336	6
Boot Head		184.342	186.556	226.736	30	29.6724	7

Minimum 185 204

MAWP: 185 kgf/cm², limited by: Right Head.

Elements Suitable for Design Internal Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Left Head

Spherical Head Design Information:					
Design Pressure	P	184.387 kgf/cm²	Int. Design Temperature	82 °C	
Material		SA-516 70	Ext. Design Temperature	82 °C	
Length		0.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1647.5 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	3137.000 mm	Actual Stress	1550.6 kgf/cm²	
Spec. Min. Thk t _e , or s _a		94.0000 mm	Spec. Nominal Thk.	102.000 mm	
Surface Area		175336.7 cm²	SG of Contents	0.658	
Element Volume		8081894.0 cm³	Weight of Contents	651.4 kgf	
Empty Weight		13031.9 kgf	Operating Weight	13683.3 kgf	
			Comp. All. Ope. Stress	1248.5 kgf/cm²	

Radiography Information:				
...		...		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 39.213 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 19.606 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:

$$= D/2(\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 3143.35/2(\exp(0.5 \cdot 184.387/(1647.49 \cdot 1.0)) - 1) + 3.175 + 0.0$$

$$= 90.4586 + 3.175 + 0.0 = 93.6336 \text{ mm}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 2 \cdot 1647.5 \cdot 1.0 \cdot \ln((2 \cdot 90.825 + 3143.35) / 3143.35) - 0.045$$

$$= 185.114 - 0.045 = 185.0687 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2 \cdot 1784.41 \cdot 1.0 \cdot \ln((2 \cdot 94.0 + 3137.0) / 3137.0)$$

$$= 207.7155 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 184.387 \cdot 3143.35^2 / (1.0 (3325.0^2 - 3143.35^2))$$

$$= 1550.557 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 184.387 / (1.0 \cdot \ln((2 \cdot 90.825 + 3143.35) / 3143.35))$$

$$= 1641.024 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 4.724 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b/(D_f - 2t))$ 40.958 %

Where D_b is the blank diameter for one piece doubly curved surfaces [Db]:



= square root(4 • Head Surface Area / 3.14159)
= square root(4 • 175336.7/3.14159)
= 4724.888 mm

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 39.213 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 19.606 mm

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	184.387 kgf/cm²	Int. Design Temperature	82 °C	
Material		SA-516 70	Ext. Design Temperature	82 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1647.5 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	3048.000 mm	Actual Stress	1548.7 kgf/cm²	
Spec. Min. Thk t _e , or s _a		185.0000 mm	Spec. Nominal Thk.	185.000 mm	
Surface Area		322138.9 cm²	SG of Contents	0.658	
Element Volume		21889810.0 cm³	Weight of Contents	2425.5 kgf	
Empty Weight		43688.9 kgf	Operating Weight	46114.4 kgf	
			Comp. All. Ope. Stress	1251.5 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3054.35 \cdot (\exp(184.387/(1647.49 \cdot 1.0)) - 1) + 3.175 + 0.0$
 $= 180.8537 + 3.175 + 0.0 = 184.0287 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1647.49 \cdot 1.0 \cdot \ln((2 \cdot 181.825 + 3054.35) / 3054.35) - 0.045$
 $= 185.324 - 0.045 = 185.2788 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1784.41 \cdot 1.0 \cdot \ln((2 \cdot 185.0 + 3048.0) / 3048.0)$
 $= 204.4397 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 184.387 \cdot 3054.35 / (1.0(3418.0 - 3054.35))$
 $= 1548.6980 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 184.387 / (1.0 \cdot \ln((2 \cdot 181.825 + 3054.35) / 3054.35))$
 $= 1639.1636 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) = 5.722 \%$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was



specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	184.387 kgf/cm²	Int. Design Temperature	82 °C	
Material		SA-516 70	Ext. Design Temperature	82 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1647.5 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	3048.000 mm	Actual Stress	1548.7 kgf/cm²	
Spec. Min. Thk t,e, or sa		185.0000 mm	Spec. Nominal Thk.	185.000 mm	
Surface Area		322138.9 cm²	SG of Contents	0.658	
Element Volume		21889810.0 cm³	Weight of Contents	2425.5 kgf	
Empty Weight		43688.9 kgf	Operating Weight	46114.4 kgf	
			Comp. All. Ope. Stress	1251.5 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3054.35 \cdot (\exp(184.387/(1647.49 \cdot 1.0)) - 1) + 3.175 + 0.0$
 $= 180.8537 + 3.175 + 0.0 = 184.0287 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1647.49 \cdot 1.0 \cdot \ln((2 \cdot 181.825 + 3054.35) / 3054.35) - 0.045$
 $= 185.324 - 0.045 = 185.2788 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1784.41 \cdot 1.0 \cdot \ln((2 \cdot 185.0 + 3048.0) / 3048.0)$
 $= 204.4397 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 184.387 \cdot 3054.35 / (1.0(3418.0 - 3054.35))$
 $= 1548.6980 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 184.387 / (1.0 \cdot \ln((2 \cdot 181.825 + 3054.35) / 3054.35))$
 $= 1639.1636 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.722 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was



specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	184.387 kgf/cm²	Int. Design Temperature	82 °C	
Material		SA-516 70	Ext. Design Temperature	82 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1647.5 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	3048.000 mm	Actual Stress	1548.7 kgf/cm²	
Spec. Min. Thk t _e , or s _a		185.0000 mm	Spec. Nominal Thk.	185.000 mm	
Surface Area		322138.9 cm²	SG of Contents	0.658	
Element Volume		21889810.0 cm³	Weight of Contents	2425.5 kgf	
Empty Weight		43688.9 kgf	Operating Weight	46114.4 kgf	
			Comp. All. Ope. Stress	1251.5 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3054.35 \cdot (\exp(184.387/(1647.49 \cdot 1.0)) - 1) + 3.175 + 0.0$
 $= 180.8537 + 3.175 + 0.0 = 184.0287 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1647.49 \cdot 1.0 \cdot \ln((2 \cdot 181.825 + 3054.35) / 3054.35) - 0.045$
 $= 185.324 - 0.045 = 185.2788 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1784.41 \cdot 1.0 \cdot \ln((2 \cdot 185.0 + 3048.0) / 3048.0)$
 $= 204.4397 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 184.387 \cdot 3054.35 / (1.0(3418.0 - 3054.35))$
 $= 1548.6980 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 184.387 / (1.0 \cdot \ln((2 \cdot 181.825 + 3054.35) / 3054.35))$
 $= 1639.1636 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.722 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was



specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:			
Design Pressure P	184.387 kgf/cm ²	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	90.600 cm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.1750 mm	Allowable Stress (ope) S	1647.5 kgf/cm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	1784.4 kgf/cm ²
Inside Diameter Di	3048.000 mm	Actual Stress	1548.7 kgf/cm ²
Spec. Min. Thk t _e , or s _a	185.0000 mm	Spec. Nominal Thk.	185.000 mm
Surface Area	97286.0 cm ²	SG of Contents	0.658
Element Volume	6610723.0 cm ³	Weight of Contents	732.6 kgf
Empty Weight	13194.0 kgf	Operating Weight	13926.6 kgf
		Comp. All. Ope. Stress	1251.5 kgf/cm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 3054.35 \cdot (\exp(184.387/(1647.49 \cdot 1.0)) - 1) + 3.175 + 0.0$
 $= 180.8537 + 3.175 + 0.0 = 184.0287 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1647.49 \cdot 1.0 \cdot \ln((2 \cdot 181.825 + 3054.35) / 3054.35) - 0.045$
 $= 185.324 - 0.045 = 185.2788 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1784.41 \cdot 1.0 \cdot \ln((2 \cdot 185.0 + 3048.0) / 3048.0)$
 $= 204.4397 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 184.387 \cdot 3054.35 / (1.0(3418.0 - 3054.35))$
 $= 1548.6980 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 184.387 / (1.0 \cdot \ln((2 \cdot 181.825 + 3054.35) / 3054.35))$
 $= 1639.1636 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) = 5.722 \%$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was



specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Right Head

Spherical Head Design Information:					
Design Pressure	P	184.387 kgf/cm²	Int. Design Temperature	82 °C	
Material		SA-516 70	Ext. Design Temperature	82 °C	
Length		0.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1647.5 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	3137.000 mm	Actual Stress	1550.6 kgf/cm²	
Spec. Min. Thk t _e , or s _a		94.0000 mm	Spec. Nominal Thk.	102.000 mm	
Surface Area		175336.7 cm²	SG of Contents	0.658	
Element Volume		8081894.0 cm³	Weight of Contents	651.4 kgf	
Empty Weight		13031.9 kgf	Operating Weight	13683.3 kgf	
			Comp. All. Ope. Stress	1251.5 kgf/cm²	

Radiography Information:				
...		...		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 39.213 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 19.606 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:

$$= D/2(\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 3143.35/2(\exp(0.5 \cdot 184.387/(1647.49 \cdot 1.0)) - 1) + 3.175 + 0.0$$

$$= 90.4586 + 3.175 + 0.0 = 93.6336 \text{ mm}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 2 \cdot 1647.5 \cdot 1.0 \cdot \ln((2 \cdot 90.825 + 3143.35) / 3143.35) - 0.045$$

$$= 185.114 - 0.045 = 185.0687 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2 \cdot 1784.41 \cdot 1.0 \cdot \ln((2 \cdot 94.0 + 3137.0) / 3137.0)$$

$$= 207.7155 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 184.387 \cdot 3143.35^2 / (1.0 (3325.0^2 - 3143.35^2))$$

$$= 1550.557 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 184.387 / (1.0 \cdot \ln((2 \cdot 90.825 + 3143.35) / 3143.35))$$

$$= 1641.024 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 4.724 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b/(D_f - 2t))$ 40.958 %

Where D_b is the blank diameter for one piece doubly curved surfaces [Db]:



= square root(4 • Head Surface Area / 3.14159)
= square root(4 • 175336.7/3.14159)
= 4724.888 mm

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 39.213 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 19.606 mm

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Boot Head

Elliptical Head Design Information:					
Design Pressure	P	184.342 kgf/cm²	Int. Design Temperature	82 °C	
Material		SA-516 70	Ext. Design Temperature	82 °C	
Length		10.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1647.5 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	508.000 mm	Actual Stress	1627.9 kgf/cm²	
Spec. Min. Thk t _e , or s _a		30.0000 mm	Spec. Nominal Thk.	36.000 mm	
Surface Area		5468.7 cm²	SG of Contents	0.000	
Element Volume		37428.9 cm³	Weight of Contents	0.0 kgf	
Empty Weight		146.5 kgf	Operating Weight	146.5 kgf	
			Comp. All. Ope. Stress	1248.5 kgf/cm²	
			Aspect Ratio	2.00	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.350 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.175 mm

Warning: L/t out of range - see 4.3.6.1(b)!

Warning: 20 ≤ L/t ≤ 2000; Actual = 17.162; This requirement does not apply to heads designed as dished bolted covers per Part 4.7.

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:
 tr = 26.4974 mm

$$\begin{aligned}
 t &= tr + ci + co \\
 &= 26.4974 + 3.175 + 0.0 \\
 &= 29.6724 \text{ mm - see below for the derivation}
 \end{aligned}$$

For Elliptical heads: Compute [h, k, r,] per 4.3.7.1

$$\begin{aligned}
 h &= D/(2 \cdot \text{Aspect Ratio}) \\
 &= 508.0/(2 \cdot 2.0) \\
 &= 127.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 k &= D/(2 \cdot h) \\
 &= 508.0/(2 \cdot 127.0) \\
 &= 2.000
 \end{aligned}$$

$$\begin{aligned}
 r &= D(0.5/k - 0.08) + ca \\
 &= 508.0(0.5/2.0 - 0.08) + 3.175 \\
 &= 89.535 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 L &= D(0.44 \cdot k + 0.02) + ca \\
 &= 508.0(0.44 \cdot 2.0 + 0.02) + 3.175 \\
 &= 460.375 \text{ mm}
 \end{aligned}$$

Step 3: Geometric Constants [βTh, φTh, Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: t = 26.4974 mm, P = 184.3421 kgf/cm²



Angle [βTh]:

$$\begin{aligned} &= \arccos[(0.5 \cdot D - r)/(L - r)] \\ &= \arccos[(0.5 \cdot 514.35 - 89.535)/(460.375 - 89.535)] \\ &= 1.1017 \text{ radians} - \text{Equation 4.3.8} \end{aligned}$$

$$\begin{aligned} \phi Th &= \sqrt{(L \cdot t) / r} \\ &= \sqrt{(460.375 \cdot 26.497) / 89.535} \\ &= 1.2336 \text{ radians} - \text{Equation 4.3.9} \end{aligned}$$

$\phi Th > \beta Th$, Thus:

$$Rth = 0.5 \cdot D = 0.5 \cdot 514.35 = 257.175 \text{ mm} - \text{Equation 4.3.11}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [C1 and C2]:

$r/D > 0.08$, Thus:

$$\begin{aligned} C1 &= 0.692(r/D) + 0.605 \\ &= 0.692(89.535/514.35) + 0.605 \\ &= 0.7255 - \text{Equation 4.3.13} \end{aligned}$$

$$\begin{aligned} C2 &= 1.46 - 2.6(r/D) = 1.46 - 2.6(89.535/514.35) \\ &= 1.0074, \text{Equation 4.3.15} \end{aligned}$$

Step 5: Pressure to produce knuckle elastic buckling [P_{eth}]:

$$\begin{aligned} &= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r)) \\ &= 0.725 \cdot 2031466 \cdot 26.497^2 / (1.007 \cdot 257.175(0.5 \cdot 257.175 - 89.535)) \\ &= 102269.91 \text{ kgf/cm}^2, \text{Equation 4.3.16} \end{aligned}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [P_y]:

$$C3 = S_y = 2473.717 \text{ kgf/cm}^2, \text{Not in the creep range} - 4.3.6.1(f)$$

$$\begin{aligned} P_y &= C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1)) \\ &= 2473.717 \cdot 26.497 / (1.007 \cdot 257.175(0.5 \cdot 257.175 / 89.535 - 1)) \\ &= 580.048 \text{ kgf/cm}^2, \text{Equation 4.3.17} \end{aligned}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and P_{ck}]:

$$G = P_{eth}/P_y = 102269.906/580.048 = 176.313$$

$G > 1$, Thus:

Buckling Pressure [P_{ck}]:

$$\begin{aligned} &= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) / \\ &\quad (1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) P_y \\ &= ((0.77508 \cdot 176.313 - 0.20354 \cdot 176.313^2 + 0.019274 \cdot 176.313^3) / \\ &\quad (1 + 0.19014 \cdot 176.313 - 0.089534 \cdot 176.313^2 + 0.0093965 \cdot 176.313^3)) \cdot 580.048 \\ &= 1183.22 \text{ kgf/cm}^2, \text{Equation 4.3.19} \end{aligned}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [P_{ak}]:

$$= P_{ck}/1.5 = 1183.22/1.5 = 788.81 \text{ kgf/cm}^2 - \text{Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [P_{ac}]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L/t + 0.5) \\ &= 2 \cdot 1647.495 \cdot 1.0 / (460.375/26.497 + 0.5) \\ &= 184.34 \text{ kgf/cm}^2, \text{Equation 4.3.22} \end{aligned}$$

Step 10: Compute final pressure [P_a]:

$$\begin{aligned} &= \min(P_{ak}, P_{ac}) = \min(788.811, 184.342) \\ &= 184.34 \text{ kgf/cm}^2, \text{Equation 4.3.23: (same as original design pressure)} \end{aligned}$$

Allowable pressure in Crown at given thickness [P_{ac}]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L/t + 0.5) \\ &= 2 \cdot 1647.495 \cdot 1.0 / (460.375/26.825 + 0.5) \\ &= 186.556 \text{ kgf/cm}^2 \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - P_{liq} = 186.5564 - 0.0 = 186.5564 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 226.7356 \text{ kgf/cm}^2$$

Computed Membrane Stress Intensity at Design Pressure [S_m]:

$$= 1627.9 \text{ kgf/cm}^2$$



Required Thickness of Straight Flange = 33.623 mm

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 25.872 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b/(D_f - 2t))$ 49.628 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

= square root($4 \cdot \text{Head Surface Area} / 3.14159$)
 = square root($4 \cdot 5468.7 / 3.14159$)
 = 834.440 mm

MDMT Calculations in the Knuckle Portion:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 13 °C
 User entered Minimum Design Temperature : -33 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 17 °C
 Temperature reduction per Graph 3.12 : 4 °C
 The computed stress ratio is : 0.914

The computed thickness ratio [sr]:
 = $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 = $(24.509 \cdot 1.0) / (30.0 - 3.175 - 0.0)$
 = 0.914

Warning: Impact Testing Is Mandatory!

MDMT Calculations in the Head Straight Flange:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 15 °C
 User entered Minimum Design Temperature : -33 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 21 °C
 Temperature reduction per Graph 3.12 : 6 °C
 The computed stress ratio is : 0.856

The computed thickness ratio [sr]:
 = $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 = $(28.105 \cdot 1.0) / (36.0 - 3.175 - 0.0)$
 = 0.856

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	185.069 kgf/cm ²
Governing Thk.	36.000 mm	Min. Des. Mtl. Temp.	-33 °C
Stress/Thk. Ratio	0.856	Basic MDMT	21 °C
		Temperature Difference	6 °C
		Computed Min. Temp.	15 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.350 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.175 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 185.0687(1.077)$
 $= 249.226 \text{ kgf/cm}^2$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Left Head	2143.7	2518.6	0.851	249.54
Shell-1	2178.0	2518.6	0.865	249.53
Shell-2	2178.0	2518.6	0.865	249.53
Shell-3	2178.0	2518.6	0.865	249.53
Shell-4	2178.0	2518.6	0.865	249.53
Right Head	2143.7	2518.6	0.851	249.54
Boot Head	1964.4	2518.6	0.780	249.60

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Nozzle_A_24"	Nozzle	1407.00	1306.00	1.077
Nozzle_B_24"	Nozzle	1407.00	1306.00	1.077
BOOT	Nozzle	1407.00	1306.00	1.077
L2_2"	Nozzle	1407.00	1306.00	1.077
L4_2"	Nozzle	1407.00	1306.00	1.077
L6_2"	Nozzle	1407.00	1306.00	1.077
L8_2"	Nozzle	1407.00	1306.00	1.077
K9_2"	Nozzle	1407.00	1306.00	1.077
K10_2"	Nozzle	1407.00	1306.00	1.077
L1_2"	Nozzle	1407.00	1306.00	1.077
L3_2"	Nozzle	1407.00	1306.00	1.077
L5_2"	Nozzle	1407.00	1306.00	1.077
L7_2"	Nozzle	1407.00	1306.00	1.077
MH-1_24"	Nozzle	1407.00	1306.00	1.077
Nozzle_C_10"	Nozzle	1407.00	1306.00	1.077

Minimum

1.077

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Left Head	1784.41	1647.49	1.083
Shell-1	1784.41	1647.49	1.083
Shell-2	1784.41	1647.49	1.083
Shell-3	1784.41	1647.49	1.083
Shell-4	1784.41	1647.49	1.083
Right Head	1784.41	1647.49	1.083
Boot Head	1784.41	1647.49	1.083

Minimum

1.083

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
Nozzle_A_24"	1600.62	1986.00	0.806
Nozzle_B_24"	1600.62	1986.00	0.806
BOOT	1561.92	1986.00	0.786



L2_2"	751.68	1986.00	0.378
L4_2"	751.68	1986.00	0.378
L6_2"	751.68	1986.00	0.378
L8_2"	751.68	1986.00	0.378
K9_2"	853.61	1986.00	0.430
K10_2"	853.61	1986.00	0.430
L1_2"	498.68	1986.00	0.251
L3_2"	498.68	1986.00	0.251
L5_2"	498.68	1986.00	0.251
L7_2"	498.68	1986.00	0.251
MH-1_24"	1600.62	1986.00	0.806
Nozzle_C_10"	1456.62	1986.00	0.733

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	3325	90.825	No Calc	No Calc
20	30	1095.17	3418	181.825	No Calc	No Calc
30	40	1095.17	3418	181.825	No Calc	No Calc
40	50	1095.17	3418	181.825	No Calc	No Calc
50	60	1095.17	3418	181.825	No Calc	No Calc
60	70	No Calc	3325	90.825	No Calc	No Calc
70	80	No Calc	568	26.825	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	94	No Calc	...	119.529
20	30	185	No Calc	...	116.236
30	40	185	No Calc	...	116.236
40	50	185	No Calc	...	116.236
50	60	185	No Calc	...	116.236
60	70	94	No Calc	...	119.529
70	80	30	No Calc	...	128.296

Minimum

116

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	1095.17	No Calc	No Calc	No Calc
30	40	1095.17	No Calc	No Calc	No Calc
40	50	1095.17	No Calc	No Calc	No Calc
50	60	1095.17	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	13031.9	8081894	12650.8	8131072	...
20	30	43688.9	21889810	42981.2	21981116	...
30	40	43688.9	21889810	42981.2	21981116	...
40	50	43688.9	21889810	42981.2	21981116	...
50	60	13194	6610723	12980.3	6638297	...
60	70	13031.9	8081894	12650.8	8131072	...
70	80	146.508	37428.9	133.587	38590.3	...
Total		170470	88481368.00	167359	88882384.00	0

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Plat	5888.69	495.4	173.87	...	PLATFORM
10	Liqd	651.356	-52.2833	122.55	...	Liquid: 10
10	Insl	360.231	-78.425	...	...	Ins: 10
10	Nozl	1265.44	-134.915	...	-80	Nozzle_A_24"
10	Nozl	1265.44	-134.915	...	80	Nozzle_B_24"
10	Wght	1600	-134.915	...	...	cut-back
20	Sadl	1129.85	198.1	172.35	...	Left Saddle
20	Liqd	2425.5	150	118.1	...	Liquid: 20
20	Insl	630.09	150	...	...	Ins: 20
20	Nozl	1908.25	100	-177.8	...	BOOT
20	Nozl	47.7371	20	-154.94	...	L2_2"
20	Nozl	47.7371	20	-145.596	-52.9926	L4_2"
20	Nozl	47.7371	20	-99.5935	-118.691	L6_2"
20	Nozl	47.7371	20	-52.9926	-145.596	L8_2"
20	Nozl	93.5382	150	154.652	...	K9 2"
20	Nozl	93.5382	200	154.652	...	K10 2"
20	Wght	39.9993	198.1	...	...	6"pipe support
20	Wght	39.9993	100	-140	...	Anti-Swirl Baff
20	Wght	9999.82	150	...	...	Misc weight
20	Wght	1419.97	100	...	170	Trunnion1
20	Wght	1419.97	100	...	-170	Trunnion2
20	Wght	209.996	198.1	-200	...	Fire proofing
30	Liqd	2425.5	150	118.1	...	Liquid: 30
30	Insl	630.09	150	...	...	Ins: 30
30	Nozl	85.4959	50.0005	...	154.94	L1_2"
30	Nozl	85.4959	100	...	154.94	L3_2"
30	Nozl	85.4959	150	...	154.94	L5_2"
30	Nozl	85.4959	200	...	154.94	L7_2"
30	Wght	199.996	149.999	100	...	demister-1
30	Wght	99.9982	149.999	90	...	demister-2
30	Wght	4799.92	149.999	...	...	Pipe+Expander
30	Wght	699.988	149.999	...	10	Shroud+Spt Clea
30	Wght	49.9991	149.999	90	...	Demister Grid1
30	Wght	49.9991	149.999	80	...	Demister Grid2
30	Wght	299.995	249.999	120	...	Distributor pla
30	Wght	149.997	200	...	...	Side Baffles
40	Sadl	1129.85	192.5	172.35	...	Right Saddle
40	Liqd	2425.5	150	118.1	...	Liquid: 40
40	Insl	630.09	150	...	...	Ins: 40
40	Wght	39.9993	192.5	...	...	6" pipe support
40	Wght	100	150	...	...	Ins Cleat
40	Wght	209.996	192.501	-200	...	Fire Proofing
40	Wght	180	150	...	...	Platform Cleat
50	Liqd	732.554	45.3	118.1	...	Liquid: 50
50	Insl	190.287	45.3	...	...	Ins: 50
50	Wght	1419.97	45.0005	...	170	Trunnion3
50	Wght	1419.97	45.0005	...	-170	Trunnion4



60	Liqd	651.356	52.2833	122.55	...	Liquid: 60
60	Insl	360.231	78.425	...	...	Ins: 60
60	Nozl	4440.02	156.85	...	...	MH-1_24"
60	Wght	199.996	156.85	...	...	Manway Davit
60	Wght	1600	156.85	...	...	cut back
70	Insl	27.984	...	-1.35	...	Ins: 70
70	Nozl	86.306	...	-36.2	...	Nozzle_C_10"

Total Weight of Each Detail Type:

Saddles	2259.7
Platforms	5888.7
Liquid	9311.8
Insulation	2829.0
Nozzles	9685.5
Weights	26249.6

Sum of the Detail Weights 56224.2 kg

Note:

Because some nozzles were attached to a head or flange, their weights include the attached head or flange.

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	170324.4	170324.4	170324.4	170324.4	170324.4	170324.4
Saddles	2259.7	2259.7	2259.7	2259.7	2259.7	2259.7
Nozzles	9685.5	9685.5	9685.5	9685.5	9685.5	9685.5
Wld Weights	25229.6	25229.6	25229.6	25229.6	25229.6	25229.6
Platforms	...	...	5888.7	5888.7	5888.7	5888.7
Insulation	...	...	2829.0	2829.0	2829.0	2829.0
Empty Weights	...	...	...	1020.0	1020.0	...
Test Weights	...	1020.0	...	...	...	...
Ope Weights	...	...	...	...	...	1020.0
Ope. Liquid	...	...	...	...	...	9311.8
Test Liquid	...	88441.4	...	...	...	...
Totals	207499.2	296960.7	216216.9	217236.9	217236.9	226548.7

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	207499.2 kg
Shop Test Wt.	- Fabricated weight + water (full)	296960.7 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	216216.9 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	217236.9 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	217236.9 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	226548.7 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	223449.7 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	175337	154578
20	30	322139	287267
30	40	322139	287267
40	50	322139	287267
50	60	97286	86754.7
60	70	175337	154578
70	80	5468.65	4393.31



Total	1419844.9	1262106.6 cm ²
Total	1528.3	1358.5 ft ²

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
L2_2"	243.77	260.14	82	1500	GR 1.1	...	...	...	...
L4_2"	243.77	260.14	82	1500	GR 1.1	...	...	...	...
L6_2"	243.77	260.14	82	1500	GR 1.1	...	...	...	...
L8_2"	243.77	260.14	82	1500	GR 1.1	...	...	...	...
L1_2"	243.77	260.14	82	1500	GR 1.1	...	...	...	...
L3_2"	243.77	260.14	82	1500	GR 1.1	...	...	...	...
L5_2"	243.77	260.14	82	1500	GR 1.1	...	...	...	...
L7_2"	243.77	260.14	82	1500	GR 1.1	...	...	...	...
MH-1_24"	243.77	260.14	82	1500	GR 1.1	...	...	...	...
Min Rating	243.766	260.136, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 cm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
= $\min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$
= $\min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.988)/(1 + 1.7 \cdot 3.4 \cdot 0.228)))$
= $\min(0.85, 0.918)$
= 0.850

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.500
Structure Height to Diameter ratio	0.045

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
Exposure Category: C from Table 26.11-1
α = 9.5: Zg = 27432.0 cm

Effective Height [z]:
= Centroid Height + Vessel Base Elevation
= 213.4 + 0.0 = 213.4 cm
= 7.001 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
Because z (7.001 ft.) < 15 ft.
= $2.01(15/Zg)^{2/\alpha}$
= $2.01(15/900.0)^{2/9.5}$
= 0.849

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
K1 = 0, K2 = 0, K3 = 0



Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.119 \text{ kgf/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.5 \cdot 50.167$$

$$= 341.1 \text{ lbs. } [154.7 \text{ kgf}]$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
Left Head	213.4	0.000	0.000	0.000	0.849	1.000	78.119
Shell-1	213.4	0.000	0.000	0.000	0.849	1.000	78.119
Shell-2	213.4	0.000	0.000	0.000	0.849	1.000	78.119
Shell-3	213.4	0.000	0.000	0.000	0.849	1.000	78.119
Shell-4	213.4	0.000	0.000	0.000	0.849	1.000	78.119
Right Head	213.4	0.000	0.000	0.000	0.849	1.000	78.119
Boot Head	234.0	0.000	0.000	0.000	0.849	1.000	78.119

Platform Load Calculations

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf	Cf
PLATFORM	111680.01	213.40	78.12	872.43	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
cut-back	0.00	213.40	78.12	0.00
6"pipe suppor	0.00	213.40	78.12	0.00
Anti-Swirl Ba	0.00	0.00	78.12	0.00
Misc weight	0.00	213.40	78.12	0.00
Trunnion1	0.00	213.40	78.12	0.00
Trunnion2	0.00	213.40	78.12	0.00
Fire proofing	0.00	0.00	78.12	0.00
demister-1	0.00	313.40	78.12	0.00
demister-2	0.00	303.40	78.12	0.00
Pipe+Expander	0.00	213.40	78.12	0.00
Shroud+Spt Cl	0.00	213.40	78.12	0.00
Demister Grid	0.00	303.40	78.12	0.00
Demister Grid	0.00	293.40	78.12	0.00
Distributor p	0.00	333.40	78.12	0.00
Side Baffles	0.00	213.40	78.12	0.00
6" pipe suppo	0.00	213.40	78.12	0.00
Ins Cleat	0.00	213.40	78.12	0.00
Fire Proofing	0.00	0.00	78.12	0.00
Platform Clea	0.00	213.40	78.12	0.00
Trunnion3	0.00	213.40	78.12	0.00
Trunnion4	0.00	213.40	78.12	0.00
Manway Davit	0.00	213.40	78.12	0.00
cut back	0.00	213.40	78.12	0.00

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
------	----	----------------------	------------------------	---------------------------------	--	-----------------------------



10	20	213.4	413.4	46606.9	78.1189	1129.88
20	30	213.4	424.56	127368	78.1189	465.155
30	40	213.4	424.56	127368	78.1189	465.155
40	50	213.4	424.56	127368	78.1189	465.155
50	60	213.4	424.56	38465.1	78.1189	140.477
60	70	213.4	413.4	46606.9	78.1189	170.211
70	80	234.042	94.56	2928.54	78.1189	10.6952

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(12.14730.75)$
 = 0.130 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.13$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.157 \cdot 221190 \cdot 2.0$
 = 20809.559 kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 221190$
 = 6936.5 kgf

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf
10	20	156.85	27648.8	2657.75
20	30	152.4	27648.8	2582.34
20	30	152.4	27648.8	2582.34
30	40	152.4	27648.8	2582.34
40	0	152.4	27648.8	2582.34
40	50	152.4	27648.8	2582.34



50	60	152.4	27648.8	2582.34
60	70	156.85	27648.8	2657.75

The Earthquake Weight does not include the weight of the saddles.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Shop/Field Installation Options :

Platform(s) installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	495.300 cm
Center of Gravity of the Platforms	495.400 cm
Center of Gravity of the Liquid	495.302 cm
Center of Gravity of the Insulation	491.389 cm
Center of Gravity of the Nozzles	530.134 cm
Center of Gravity of the Added Weights (Operating)	368.534 cm
Center of Gravity of the Added Weights (Empty)	368.534 cm
Center of Gravity of Bare Shell New and Cold	494.960 cm
Center of Gravity of Bare Shell Corroded	494.984 cm
Vessel CG in the Operating Condition	481.643 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	481.230 cm
Vessel CG in the Test Condition	485.250 cm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (305.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(280.0 + 1.56 \cdot \sqrt{ 1618.0874 \cdot 181.825 }, 2 \cdot 1981.0)$$

$$= 1126.1598 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1618.09	mm
Shell Thickness used in this Case	t	185.000	mm
Stiffened Vessel Length per 4.15.6	L	990.60	cm
Distance from Saddle to Vessel tangent	a1 or a	1981.00	mm
Saddle Width	b1 or b	280.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	161.71	cm
Shell Allowable Stress used in Calculation		1647.49	kgf/cm ²
Head Allowable Stress used in Calculation		1647.49	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2134.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		142854.48	kgf
Pressure used in Analysis	P	184.342	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	819.74	1647.49
Long. Stress at Bottom of Midspan	820.75	1647.49
Long. Stress at Top of Saddles	881.11	1647.49
Long. Stress at Bottom of Saddles	786.51	1647.49
Tangential Shear in Shell	28.01	1317.99
Circ. Stress at Horn of Saddle	62.21	2059.36
Circ. Compressive Stress in Shell	5.30	1647.49

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(2846.7 / 2 + 0) \cdot 2134.0 / 2780.0$$

$$= 3277.8 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(1510.05, 0) \cdot 2134.0 / 5944.0005$$

$$= 542.1 \text{ kgf}$$

Saddle Reaction Force due to Earthquake FI [Fsl]:

$$= \max(F_{l, \text{eq}}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(20809.56, 0) \cdot 2134.0 / 5944.0005$$

$$= 7471.0 \text{ kgf}$$



Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(20810/2 + 0) \cdot 2134.0/2780.0$
 $= 23960.9\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 118894 + \max(542, 3278, 7471, 23961)$
 $= 142854.5\ kgf$

Longitudinal Wind Force [FI]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766.08(3.416 + \pi/4(3.538 \cdot 1.2)^2)$
 $= 14808.577\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	143984.33	kgf
Transverse Shear Load	10404.78	kgf
Longitudinal Shear Load	20809.56	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= Baseplate\ Length$
 $= 2780.000\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -142854 \cdot 198.1 [1 - (1 - 198.1/990.6 + (161.809^2 - 161.709^2) / (2 \cdot 198.1 \cdot 990.6)) / (1 + (4 \cdot 161.71) / (3 \cdot 990.6))]$
 $= -97043.7\ kgf \cdot m.$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 142854 \cdot 991 / 4 ((1 + 2(162^2 - 162^2)/(991^2)) / (1 + (4 \cdot 162) / (3 \cdot 991)) - 4 \cdot 198 / 991)$
 $= 7565.1\ kgf \cdot m.$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$
 $= 184.342 \cdot 1618.087 / (2 \cdot 181.825) - 7565.1 / (\pi \cdot 1618.1^2 \cdot 181.825)$
 $= 819.74\ kgf/cm^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$
 $= 184.342 \cdot 1618.087 / (2 \cdot 181.825) + 7565.1 / (\pi \cdot 1618.1^2 \cdot 181.825)$
 $= 820.75\ kgf/cm^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 $= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$
 $= 184.342 \cdot 1618.087 / (2 \cdot 181.825) - 97043.7 / (0.1066 \cdot \pi \cdot 1618.1^2 \cdot 181.825)$
 $= 881.11\ kgf/cm^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:
 $= P \cdot Rm / (2t) + M1 / (K1* \cdot \pi \cdot Rm^2 \cdot t)$
 $= 184.342 \cdot 1618.087 / (2 \cdot 181.825) + 97043.7 / (0.1923 \cdot \pi \cdot 1618.1^2 \cdot 181.825)$
 $= 786.51\ kgf/cm^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L - 2a) / (L + (4 \cdot hm / 3))$
 $= 142854(990.6 - 2 \cdot 198.1) / (990.6 + (4 \cdot 161.71 / 3))$
 $= 70396.2\ kgf$



Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 70396.18 / (1618.0874 \cdot 181.825)$$

$$= 28.01 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1618.087 \cdot 181.825)}$$

$$= 423.080 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 142854 \cdot 0.1 / (181.825(280.0 + 423.08 + 423.08))$$

$$= -5.30 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -142854 / (4 \cdot 181.825(280.0 + 423.08 + 423.08)) -$$

$$12 \cdot 0.053 \cdot 142854 \cdot 1618.087 / (990.6 \cdot 181.825^2)$$

$$= -62.21 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(280.0 + 1.56 \cdot \sqrt{(1618.087 \cdot 181.825)}, 2 \cdot 198.1)$$

$$= 1126.16 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 5944.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 5944.0(82.0 - 21.1)$$

$$= 4.338 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2780.0000	mm
Baseplate Thickness	Bpthk	30.0000	mm
Baseplate Width	Bpwid	305.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2507.4	kgf/cm ²
Height of Web at Center	Hw,c	370.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1527.175 \cdot 181.825)} + 305.0)181.825^3) / 12$$

$$= 56457.359 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 56457.359 + 2049.25(124.183 - 90.913)^2$$

$$= 79141.312 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1127.0	181.8	90.9	2049.3	18630246.0	0.791E+05
Wearplate	305.0	26.0	194.8	79.3	1544962.1	0.400E+04
Web	20.0	369.0	392.3	73.8	2895358.8	0.614E+05
BasePlate	305.0	30.0	591.8	91.5	5415199.0	0.200E+06
Totals	...	...	...	2293.9	28485768.0	0.345E+06

Distance to Centroid [C1]:



$$\begin{aligned}
 &= AY / A \\
 &= 28485768 / 2293.85 \\
 &= 124.183 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120.0 / 2 \\
 &= 120.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.204 \cdot 142854.484 \\
 &= 29073.9844 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 118.8635 \text{ kgf/cm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1504.4374 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 2134.0 - 1527.17 \cdot \sin(120.0/2) / 1.0472 \\
 &= 871.036 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 25324.5293 \text{ kgf-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 91.2225 \text{ kgf/cm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1671.5970 \text{ kgf/cm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}) } \\
 &= \sqrt{ 3(142854 + 1130)305.0 / (4 \cdot 2780.0 \cdot 1671.597) } \\
 &= 26.623 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0 / 2) (1524.0 + 185.0 + 26.0) \\
 &= 3005.108 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 3005.1079 / (6 - 1) \\
 &= 601.022 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot B_{\text{pwid}} / 2 \\
 &= 601.0215 \cdot 305.0 / 2 \\
 &= 916.558 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 143984.344 / (2780.0 \cdot 305.0) \\
 &= 16.981 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 916.6 \cdot 16.98 \\
 &= 15564.331 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of the Rib and Web [Ar]:} \\
 &= \text{Rib Area} + \text{Web Area} \\
 &= 52.0 + 60.102 \\
 &= 112.102 \text{ cm}^2
 \end{aligned}$$

$$\text{Compressive Stress [Sc]:}$$



$$= P/Ar$$

$$= 15564.3/112.1022$$

$$= 138.841 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	280.0	...	56.0	...	0.366E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 280.0 - 20.0$$

$$= 260.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/112.102$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 280.0/2$$

$$= 140.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3658.7/112.102)}$$

$$= 57.129 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2134.0 - \cos(120.0/2) (1524.0 + 185.0 + 26.0) - 30.0$$

$$= 1236.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2) (1524.0 + 185.0 + 26.0)$$

$$= 3005.108 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2780.0 - 50.8$$

$$= 2729.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (3005.108 - 2729.2) / 2$$

$$= 137.954 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(Lv^2 + Dd^2)}$$

$$= \sqrt{(1236.5^2 + 137.954^2)}$$

$$= 1244.172 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900 / 2507.4)}$$

$$= 126.693$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1244.172 / 57.129$$

$$= 21.778$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 20809.6 / (2 \cdot 2780.0) \cdot 601.022 \cdot 1244.17 / 2$$



= 1399.359 kgf-m.

Compressive Allowable, $KL/r < Cc$ (21.7784 < 126.6926) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (21.78)^2 / (2 \cdot 126.69^2)) 2507 / (5/3 + 3 \cdot (21.78) / (8 \cdot 126.69) - (21.78^3) / (8 \cdot 126.69^3))$$

$$= 1428 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 138.84/1427.54 + (1399.36 \cdot 140.0/36586676) / 1671.6$$

$$= 0.418$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	260.0	0.0	52.0	0.0	0.366E+04
Web	601.0	20.0	0.0	120.2	0.0	40.1
Totals	...	...	...	172.2	...	0.370E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/172.204$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 280.0/2$$

$$= 140.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 2134.0 - \sqrt{ (1735.0 + 26.0)^2 - (601.022/2)^2 } - 30.0$$

$$= 395.223 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 3697.4/172.204 }$$

$$= 46.337 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 395.223/46.337$$

$$= 8.529$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 20809.561 / (2 \cdot 2780.0)$$

$$= 3.743 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.743 \cdot 601.022 \cdot 395.223$$

$$= 889.040 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σB ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 889.04/264100.125$$

$$= 336.630 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ (8.5294 < 126.6926) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (8.53)^2 / (2 \cdot 126.69^2)) 2507 / (5/3 + 3 \cdot (8.53) / (8 \cdot 126.69) - (8.53^3) / (8 \cdot 126.69^3))$$

$$= 1479 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$



$$= 180.85/1478.66 + (889.04 \cdot 140.0/3697.399)/1671.6$$

$$= 0.324$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1407.16 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	75.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1671.25 kgf/cm ²
Area Available in a Single Bolt	BlArea	6.0075 cm ²
Saddle Load QO (Weight)	QO	120023.4 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	7471.0 kgf
Maximum Transverse Force	Ft	10404.8 kgf
Maximum Longitudinal Force	Fl	20809.6 kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 33148.9 \text{ kgf}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 33148.9 / (6.008 \cdot 8)$$

$$= 689.7 \text{ kgf/cm}^2. \text{ Must be less than } 844.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 10404.8 / (6.008 \cdot 8)$$

$$= 216.5 \text{ kgf/cm}^2. \text{ Must be less than } 844.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_O > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 20809.56 / (844.3 \cdot 8.0)$$

$$= 3.0809 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 2134.0 \cdot 10404.78 + 0.0$$

$$= 22203.83 \text{ kgf-m.}$$

Eccentricity (e):

$$= R_{mom} / Q_O$$

$$= 22203.83 / 120023.39$$

$$= 185.00 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$$

$$= \max[0.0, 3.0809, 0.0]$$

$$= 3.0809 \text{ cm}^2$$

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:



Warning - Distance to Saddle (a) > 0.25 * Tangent Distance (L) - 4.15.3.2

Note:

Wear Pad Width (305.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$
 $= \min(280.0 + 1.56 \cdot \sqrt{ 1618.0874 \cdot 181.825 }, 2 \cdot 7925.0)$
 $= 1126.1598 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1618.09	mm
Shell Thickness used in this Case	t	185.000	mm
Stiffened Vessel Length per 4.15.6	L	990.60	cm
Distance from Saddle to Vessel tangent	a1 or a	7925.00	mm
Saddle Width	b1 or b	280.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	161.71	cm
Shell Allowable Stress used in Calculation		1647.49	kgf/cm ²
Head Allowable Stress used in Calculation		1647.49	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2134.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		133327.50	kgf
Pressure used in Analysis	P	184.342	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	872.76	1647.49
Long. Stress at Bottom of Midspan	767.73	1647.49
Long. Stress at Top of Saddles	1374.09	1647.49
Long. Stress at Bottom of Saddles	513.27	1647.49
Tangential Shear in Shell	26.14	1317.99
Circ. Stress at Horn of Saddle	58.06	2059.36
Circ. Compressive Stress in Shell	4.95	1647.49

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(2846.7/2 + 0) \cdot 2134.0/2780.0$
 $= 3277.8 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(1510.05, 0) \cdot 2134.0/5944.0005$
 $= 542.1 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(20809.56, 0) \cdot 2134.0/5944.0005$
 $= 7471.0 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(20810/2 + 0) \cdot 2134.0/2780.0$
 $= 23960.9 \text{ kgf}$



Load Combination Results for Q + Wind or Seismic [Q]:

$$\begin{aligned} &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\ &= 109367 + \max(542, 3278, 7471, 23961) \\ &= 133327.5 \text{ kgf} \end{aligned}$$

Longitudinal Wind Force [FI]:

$$\begin{aligned} &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2)) \\ &= 1.1 \cdot 766.08(3.416 + \pi/4(3.538 \cdot 1.2)^2) \\ &= 14808.577 \text{ N} \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	134457.36	kgf
Transverse Shear Load	10404.78	kgf
Longitudinal Shear Load	20809.56	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned} &= \text{Baseplate Length} \\ &= 2780.000 \text{ mm} \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned} &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\ &= -133328 \cdot 792.5 [1 - (1 - 792.5/990.6 + (161.809^2 - 161.709^2)/(2 \cdot 792.5 \cdot 990.6))/(1 + (4 \cdot 161.71)/(3 \cdot 990.6))] \\ &= -883071.4 \text{ kgf-m.} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} &= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L) \\ &= 133328 \cdot 991/4((1 + 2(162^2 - 162^2)/(991^2))/(1 + (4 \cdot 162)/(3 \cdot 991)) - 4 \cdot 793/991) \\ &= -785439.1 \text{ kgf-m.} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\ &= 184.342 \cdot 1618.087/(2 \cdot 181.825) - 785439.1/(\pi \cdot 1618.1^2 \cdot 181.825) \\ &= 872.76 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned} &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\ &= 184.342 \cdot 1618.087/(2 \cdot 181.825) + 785439.1/(\pi \cdot 1618.1^2 \cdot 181.825) \\ &= 767.73 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned} &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\ &= 184.342 \cdot 1618.087/(2 \cdot 181.825) - 883071.4/(0.1066 \cdot \pi \cdot 1618.1^2 \cdot 181.825) \\ &= 1374.09 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned} &= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t) \\ &= 184.342 \cdot 1618.087/(2 \cdot 181.825) + 883071.4/(0.1923 \cdot \pi \cdot 1618.1^2 \cdot 181.825) \\ &= 513.27 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned} &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\ &= 133328(990.6 - 2 \cdot 792.5)/(990.6 + (4 \cdot 161.71/3)) \\ &= -65701.5 \text{ kgf} \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned} &= K2 \cdot T / (R_m \cdot t) \\ &= 1.1707 \cdot -65701.45/(1618.0874 \cdot 181.825) \\ &= -26.14 \text{ kgf/cm}^2 \end{aligned}$$



Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1618.087 \cdot 181.825)}$$

$$= 423.080 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 133328 \cdot 0.1 / (181.825(280.0 + 423.08 + 423.08))$$

$$= -4.95 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -133328 / (4 \cdot 181.825(280.0 + 423.08 + 423.08)) -$$

$$12 \cdot 0.053 \cdot 133328 \cdot 1618.087 / (990.6 \cdot 181.825^2)$$

$$= -58.06 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(280.0 + 1.56 \cdot \sqrt{(1618.087 \cdot 181.825)}, 2 \cdot 792.5)$$

$$= 1126.16 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2780.0000	mm
Baseplate Thickness	Bpthk	30.0000	mm
Baseplate Width	Bpwid	305.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2507.4	kgf/cm ²
Height of Web at Center	Hw,c	370.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1527.175 \cdot 181.825)} + 305.0)181.825^3)/12$$

$$= 56457.359 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 56457.359 + 2049.25(124.183 - 90.913)^2$$

$$= 79141.312 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1127.0	181.8	90.9	2049.3	18630246.0	0.791E+05
Wearplate	305.0	26.0	194.8	79.3	1544962.1	0.400E+04
Web	20.0	369.0	392.3	73.8	2895358.8	0.614E+05
BasePlate	305.0	30.0	591.8	91.5	5415199.0	0.200E+06
Totals	...	...	...	2293.9	28485768.0	0.345E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 28485768/2293.85$$

$$= 124.183 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$



= 0.2035

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 133327.5$$

$$= 27135.0371 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 110.9365 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1504.4374 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2134.0 - 1527.17 \cdot \sin(120.0/2) / 1.0472$$

$$= 871.036 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 23635.6348 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 85.1389 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1671.5970 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(133328 + 1130) \cdot 305.0) / (4 \cdot 2780.0 \cdot 1671.597)}$$

$$= 25.727 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1524.0 + 185.0 + 26.0)$$

$$= 3005.108 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 3005.1079 / (6 - 1)$$

$$= 601.022 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 601.0215 \cdot 305.0/2$$

$$= 916.558 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 134457.359 / (2780.0 \cdot 305.0)$$

$$= 15.858 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 916.6 \cdot 15.86$$

$$= 14534.490 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 52.0 + 60.102$$

$$= 112.102 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 14534.5/112.1022$$

$$= 129.654 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	280.0	...	56.0	...	0.366E+04



Rib dimension [D]:

$$\begin{aligned} &= \text{Saddle Width} - \text{Web Thickness} \\ &= 280.0 - 20.0 \\ &= 260.000 \text{ mm} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0.0/112.102 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \text{Saddle Width} / 2 \\ &= 280.0/2 \\ &= 140.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ &= \sqrt{(3658.7/112.102)} \\ &= 57.129 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned} &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\ &= 2134.0 - \cos(120.0/2) (1524.0 + 185.0 + 26.0) - 30.0 \\ &= 1236.500 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned} &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\ &= 2 \cdot \sin(120.0/2) (1524.0 + 185.0 + 26.0) \\ &= 3005.108 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned} &= \text{Baseplate length} - \text{clearance} \\ &= 2780.0 - 50.8 \\ &= 2729.200 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned} &= (Lw - Bd) / 2 \\ &= (3005.108 - 2729.2) / 2 \\ &= 137.954 \text{ mm} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= \sqrt{Lv^2 + Dd^2} \\ &= \sqrt{1236.5^2 + 137.954^2} \\ &= 1244.172 \text{ mm} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\ &= \sqrt{(2 \cdot \pi^2 \cdot 2038900 / 2507.4)} \\ &= 126.693 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL/r \\ &= 1 \cdot 1244.172 / 57.129 \\ &= 21.778 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned} &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\ &= 20809.6 / (2 \cdot 2780.0) \cdot 601.022 \cdot 1244.17 / 2 \\ &= 1399.359 \text{ kgf-m.} \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($21.7784 < 126.6926$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\ &= (1 - (21.78)^2 / (2 \cdot 126.69^2)) 2507 / \\ &\quad (5/3 + 3 \cdot (21.78) / (8 \cdot 126.69) - (21.78^3) / (8 \cdot 126.69^3)) \\ &= 1428 \text{ kgf/cm}^2 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned} &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\ &= 129.65 / 1427.54 + (1399.36 \cdot 140.0 / 36586676) / 1671.6 \\ &= 0.411 \end{aligned}$$



Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	260.0	0.0	52.0	0.0	0.366E+04
Web	601.0	20.0	0.0	120.2	0.0	40.1
Totals	...	...	...	172.2	...	0.370E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/172.204$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 280.0/2$$

$$= 140.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 2134.0 - \sqrt{(1735.0 + 26.0)^2 - (601.022/2)^2} - 30.0$$

$$= 395.223 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3697.4/172.204)}$$

$$= 46.337 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 395.223/46.337$$

$$= 8.529$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 20809.561/(2 \cdot 2780.0)$$

$$= 3.743 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.743 \cdot 601.022 \cdot 395.223$$

$$= 889.040 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 889.04/264100.125$$

$$= 336.630 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (8.5294 < 126.6926) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.53)^2 / (2 \cdot 126.69^2)) 2507 /$$

$$(5/3 + 3 \cdot (8.53) / (8 \cdot 126.69) - (8.53^3) / (8 \cdot 126.69^3))$$

$$= 1479 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 168.88/1478.66 + (889.04 \cdot 140.0/3697.399) / 1671.6$$

$$= 0.316$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1407.16 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	75.0 mm



Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1671.25 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.0075 cm ²
Saddle Load QO (Weight)	QO	110496.4 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	7471.0 kgf
Maximum Transverse Force	Ft	10404.8 kgf
Maximum Longitudinal Force	Fl	20809.6 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$\begin{aligned} &= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 20809.6 / (6.008 \cdot 8) \\ &= 433.0 \text{ kgf/cm}^2. \text{ Must be less than } 844.3 \text{ kgf/cm}^2. \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$\begin{aligned} &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 10404.8 / (6.008 \cdot 8) \\ &= 216.5 \text{ kgf/cm}^2. \text{ Must be less than } 844.3 \text{ kgf/cm}^2. \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned} &= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolls}) \\ &= 20809.56 / (844.3 \cdot 8.0) \\ &= 3.0809 \text{ cm}^2 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned} &= B \cdot Ft + \text{Sum of X Moments} \\ &= 2134.0 \cdot 10404.78 + 0.0 \\ &= 22203.83 \text{ kgf-m.} \end{aligned}$$

Eccentricity (e):

$$\begin{aligned} &= Rmom / QO \\ &= 22203.83 / 110496.42 \\ &= 200.95 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction} \end{aligned}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\ &= \max[0.0, 3.0809, 0.0] \\ &= 3.0809 \text{ cm}^2 \end{aligned}$$

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (305.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(280.0 + 1.56 \cdot \sqrt{ 1616.5 \cdot 185.0 }, 2 \cdot 1981.0)$$

$$= 1133.0968 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1616.50	mm
Shell Thickness used in this Case	t	185.000	mm
Stiffened Vessel Length per 4.15.6	L	990.60	cm
Distance from Saddle to Vessel tangent	a1 or a	1981.00	mm
Saddle Width	b1 or b	280.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	161.55	cm
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2134.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		158142.28	kgf
Pressure used in Analysis	P	249.388	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	1089.00	2518.57
Long. Stress at Bottom of Midspan	1090.11	2518.57
Long. Stress at Top of Saddles	1155.88	2518.57
Long. Stress at Bottom of Saddles	1052.79	2518.57
Tangential Shear in Shell	30.51	1413.05
Circ. Stress at Horn of Saddle	66.68	3777.85
Circ. Compressive Stress in Shell	5.74	2518.57

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(1423.4 / 2 + 0) \cdot 2134.0 / 2780.0$$

$$= 1638.9 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(755.03, 0) \cdot 2134.0 / 5944.0005$$

$$= 271.1 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 156503 + \max(271, 1639, 0, 0)$$

$$= 158142.3 \text{ kgf}$$



Longitudinal Wind Force [F_L]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(3.416 + \pi/4(3.538 \cdot 1.2)^2)$$

$$= 14808.577 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	159272.14	kgf
Transverse Shear Load	711.68	kgf
Longitudinal Shear Load	755.03	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2780.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -158142 \cdot 198.1 [1 - (1 - 198.1/990.6 + (161.65^2 - 161.55^2)/(2 \cdot 198.1 \cdot 990.6))/(1 + (4 \cdot 161.55)/(3 \cdot 990.6))]$$

$$= -107392.9 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 158142 \cdot 991/4((1 + 2(162^2 - 162^2)/(991^2))/(1 + (4 \cdot 162)/(3 \cdot 991)) - 4 \cdot 198/991)$$

$$= 8431.2 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 249.388 \cdot 1616.5/(2 \cdot 185.0) - 8431.2/(\pi \cdot 1616.5^2 \cdot 185.0)$$

$$= 1089.00 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 249.388 \cdot 1616.5/(2 \cdot 185.0) + 8431.2/(\pi \cdot 1616.5^2 \cdot 185.0)$$

$$= 1090.11 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 249.388 \cdot 1616.5/(2 \cdot 185.0) - 107392.9/(0.1066 \cdot \pi \cdot 1616.5^2 \cdot 185.0)$$

$$= 1155.88 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 249.388 \cdot 1616.5/(2 \cdot 185.0) + 107392.9/(0.1923 \cdot \pi \cdot 1616.5^2 \cdot 185.0)$$

$$= 1052.79 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m)/3)$$

$$= 158142(990.6 - 2 \cdot 198.1)/(990.6 + (4 \cdot 161.55)/3)$$

$$= 77943.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ₂]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 77943.43/(1616.5 \cdot 185.0)$$

$$= 30.51 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x₁, x₂]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1616.5 \cdot 185.0)}$$

$$= 426.548 \text{ mm}$$



Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$
 $= -0.7603 \cdot 158142 \cdot 0.1 / (185.0(280.0 + 426.55 + 426.55))$
 $= -5.74 \text{ kgf/cm}^2$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:
 $= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$
 $= -158142 / (4 \cdot 185.0(280.0 + 426.548 + 426.548)) -$
 $12 \cdot 0.053 \cdot 158142 \cdot 1616.5 / (990.6 \cdot 185.0^2)$
 $= -66.68 \text{ kgf/cm}^2$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(280.0 + 1.56 \cdot \sqrt{1616.5 \cdot 185.0}, 2 \cdot 198.1)$
 $= 1133.10 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2780.0000	mm
Baseplate Thickness	Bpthk	30.0000	mm
Baseplate Width	Bpwid	305.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	370.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{1524.0 \cdot 185.0}) + 305.0)^3 / 12$
 $= 59798.406 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 59798.406 + 2096.659(125.263 - 92.5)^2$
 $= 82303.992 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1133.3	185.0	92.5	2096.7	19394094.0	0.823E+05
Wearplate	305.0	26.0	198.0	79.3	1570139.8	0.424E+04
Web	20.0	369.0	395.5	73.8	2918790.2	0.623E+05
BasePlate	305.0	30.0	595.0	91.5	5444250.0	0.202E+06
Totals	...	...	...	2341.3	29327272.0	0.351E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 29327272 / 2341.259$
 $= 125.263 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120.0 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 158142.281$
 $= 32185.3867 \text{ kgf}$



Tension Stress, $St = (Fh/As)$ = 131.5839 kgf/cm²
 Allowed Stress, $Sa = 0.6 * Yield Str$ = 1590.6736 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 2134.0 - 1524.0 \cdot \sin(120.0/2) / 1.0472$
 $= 873.662 \text{ mm}$

Bending Moment, $M = Fh * d$ = 28119.1875 kgf-m.

Bending Stress, $Sb = (M * C1 / I)$ = 100.4133 kgf/cm²
 Allowed Stress, $Sa = 2/3 * Yield Str$ = 1767.4150 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + SaddleWt)BasePlateWidth / (4 * BasePlateLength * AllStress)}$
 $= \sqrt{3(158142 + 1130)305.0 / (4 * 2780.0 * 1767.415)}$
 $= 27.231 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - Saddle Angle/2) (Inside Radius + Shell Thk + Wear Plate Thk)$
 $= 2 \cdot \cos(90 - 120.0/2) (1524.0 + 185.0 + 26.0)$
 $= 3005.108 \text{ mm}$

Distance between Ribs [e]:
 $= Web Length / (Nribs - 1)$
 $= 3005.1079 / (6 - 1)$
 $= 601.022 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot Bpwid / 2$
 $= 601.0215 \cdot 305.0 / 2$
 $= 916.558 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$
 $= 159272.141 / (2780.0 \cdot 305.0)$
 $= 18.784 \text{ kgf/cm}^2$

Axial Load [P]:
 $= Ap \cdot Bp$
 $= 916.6 \cdot 18.78$
 $= 17216.902 \text{ kgf}$

Area of the Rib and Web [Ar]:
 $= Rib Area + Web Area$
 $= 52.0 + 60.102$
 $= 112.102 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 17216.9 / 112.1022$
 $= 153.582 \text{ kgf/cm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

B	D	Y	A	AY	Io	
Rib+Web	20.0	280.0	...	56.0	...	0.366E+04

Rib dimension [D]:
 $= Saddle Width - Web Thickness$
 $= 280.0 - 20.0$
 $= 260.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$



$$= 0.0/112.102$$
$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$
$$= 280.0/2$$
$$= 140.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$
$$= \sqrt{(3658.7/112.102)}$$
$$= 57.129 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$
$$= 2134.0 - \cos(120.0/2)(1524.0 + 185.0 + 26.0) - 30.0$$
$$= 1236.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$
$$= 2 \cdot \sin(120.0/2)(1524.0 + 185.0 + 26.0)$$
$$= 3005.108 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$
$$= 2780.0 - 50.8$$
$$= 2729.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd})/2$$
$$= (3005.108 - 2729.2)/2$$
$$= 137.954 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(\text{Lv}^2 + \text{Dd}^2)}$$
$$= \sqrt{(1236.5^2 + 137.954^2)}$$
$$= 1244.172 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$
$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900/2651.1)}$$
$$= 123.211$$

Slenderness ratio [KL/r]:

$$= \text{KL}/r$$
$$= 1 \cdot 1244.172/57.129$$
$$= 21.778$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$
$$= 755.0 / (2 \cdot 2780.0) \cdot 601.022 \cdot 1244.17/2$$
$$= 50.772 \text{ kgf-m.}$$

Compressive Allowable, $\text{KL}/r < \text{Cc}$ ($21.7784 < 123.2106$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot \text{Cc}) - (\text{KL}/r)^3 / (8 \cdot \text{Cc}^3))$$
$$= (1 - (21.78)^2 / (2 \cdot 123.21^2)) 2651 /$$
$$(5/3 + 3 \cdot (21.78) / (8 \cdot 123.21) - (21.78^3) / (8 \cdot 123.21^3))$$
$$= 1507 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$$
$$= 153.58/1506.53 + (50.77 \cdot 140.0/36586676) / 1767.42$$
$$= 0.113$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	260.0	0.0	52.0	0.0	0.366E+04



Web	601.0	20.0	0.0	120.2	0.0	40.1
Totals	...	...	...	172.2	...	0.370E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/172.204$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 280.0/2$$

$$= 140.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 2134.0 - \sqrt{(1735.0 + 26.0)^2 - (601.022/2)^2} - 30.0$$

$$= 395.223 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3697.4/172.204)}$$

$$= 46.337 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 395.223/46.337$$

$$= 8.529$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 755.025/(2 \cdot 2780.0)$$

$$= 0.136 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.136 \cdot 601.022 \cdot 395.223$$

$$= 32.257 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 32.257/264100.125$$

$$= 12.214 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (8.5294 < 123.2106) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.53)^2 / (2 \cdot 123.21^2)) 2651 /$$

$$(5/3 + 3 \cdot (8.53) / (8 \cdot 123.21) - (8.53)^3 / (8 \cdot 123.21^3))$$

$$= 1563 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 200.05/1562.56 + (32.26 \cdot 140.0/3697.399) / 1767.42$$

$$= 0.135$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	75.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BlArea	6.0075 cm ²
Saddle Load Q0 (Weight)	Q0	157633.2 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	271.1 kgf
Maximum Transverse Force	Ft	711.7 kgf
Maximum Longitudinal Force	Fl	755.0 kgf



Saddle Bolted to Steel Foundation

No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $755.0 / (6.008 \cdot 8)$
= 15.7 kgf/cm^2 . Must be less than 1052.3 kgf/cm^2 .

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $711.7 / (6.008 \cdot 8)$
= 14.8 kgf/cm^2 . Must be less than 1052.3 kgf/cm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolls})$
= $755.03 / (1052.29 \cdot 8.0)$
= 0.0897 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $2134.0 \cdot 711.68 + 0.0$
= 1518.73 kgf-m .

Eccentricity (e):
= R_{mom} / QO
= $1518.73 / 157633.23$
= $9.63 \text{ mm} < B_{plen}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 0.0897, 0.0]$
= 0.0897 cm^2

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

*Warning - Distance to Saddle (a) > 0.25 * Tangent Distance (L) - 4.15.3.2*

Note:
Wear Pad Width (305.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
= $\min(280.0 + 1.56 \cdot \sqrt{1616.5 \cdot 185.0}, 2 \cdot 7925.0)$
= 1133.0968 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1616.50	mm
Shell Thickness used in this Case	t	185.000	mm
Stiffened Vessel Length per 4.15.6	L	990.60	cm
Distance from Saddle to Vessel tangent	a1 or a	7925.00	mm
Saddle Width	b1 or b	280.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	161.55	cm
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²



Circumferential Efficiency in Plane of Saddle		1.00
Circumferential Efficiency at Mid-Span		1.00
Distance from Saddle Base to Centerline	B	2134.00 mm
Coefficient of Friction	mu	0.30

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		148700.72 kgf
Pressure used in Analysis	P	249.388 kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	1147.23	2518.57
Long. Stress at Bottom of Midspan	1031.88	2518.57
Long. Stress at Top of Saddles	1697.83	2518.57
Long. Stress at Bottom of Saddles	752.41	2518.57
Tangential Shear in Shell	28.69	1413.05
Circ. Stress at Horn of Saddle	62.70	3777.85
Circ. Compressive Stress in Shell	5.39	2518.57

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (1423.4/2 + 0) \cdot 2134.0/2780.0$
 $= 1638.9 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(755.03, 0) \cdot 2134.0/5944.0005$
 $= 271.1 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 147062 + \max(271, 1639, 0, 0)$
 $= 148700.7 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08 (3.416 + \pi/4 (3.538 \cdot 1.2)^2)$
 $= 14808.577 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	149830.56 kgf
Transverse Shear Load	711.68 kgf
Longitudinal Shear Load	755.03 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 2780.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -148701 \cdot 792.5 [1 - (1 - 792.5/990.6 + (161.65^2 - 161.55^2)/(2 \cdot 792.5 \cdot 990.6))/(1 + (4 \cdot 161.55)/(3 \cdot 990.6))]$
 $= -984859.3 \text{ kgf-m.}$



Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L)$$

$$= 148701 \cdot 991 / 4 \cdot ((1 + 2(162^2 - 162^2) / (991^2)) / (1 + (4 \cdot 162) / (3 \cdot 991)) - 4 \cdot 793 / 991)$$

$$= -875950.3 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 249.388 \cdot 1616.5 / (2 \cdot 185.0) - 875950.3 / (\pi \cdot 1616.5^2 \cdot 185.0)$$

$$= 1147.23 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 249.388 \cdot 1616.5 / (2 \cdot 185.0) + 875950.3 / (\pi \cdot 1616.5^2 \cdot 185.0)$$

$$= 1031.88 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 249.388 \cdot 1616.5 / (2 \cdot 185.0) - 984859.3 / (0.1066 \cdot \pi \cdot 1616.5^2 \cdot 185.0)$$

$$= 1697.83 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 249.388 \cdot 1616.5 / (2 \cdot 185.0) + 984859.3 / (0.1923 \cdot \pi \cdot 1616.5^2 \cdot 185.0)$$

$$= 752.41 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 148701(990.6 - 2 \cdot 792.5) / (990.6 + (4 \cdot 161.55 / 3))$$

$$= -73290.0 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot -73289.97 / (1616.5 \cdot 185.0)$$

$$= -28.69 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1616.5 \cdot 185.0)}$$

$$= 426.548 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 148701 \cdot 0.1 / (185.0(280.0 + 426.55 + 426.55))$$

$$= -5.39 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -148701 / (4 \cdot 185.0(280.0 + 426.548 + 426.548)) - 12 \cdot 0.053 \cdot 148701 \cdot 1616.5 / (990.6 \cdot 185.0^2)$$

$$= -62.70 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(280.0 + 1.56 \cdot \sqrt{(1616.5 \cdot 185.0)}, 2 \cdot 792.5)$$

$$= 1133.10 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2780.0000	mm
Baseplate Thickness	Bpthk	30.0000	mm
Baseplate Width	Bpwid	305.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	370.0	mm
Friction Coefficient	mu	0.300	



Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1524.0 \cdot 185.0)} + 305.0)^3) / 12$$

$$= 59798.406 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 59798.406 + 2096.659 (125.263 - 92.5)^2$$

$$= 82303.992 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1133.3	185.0	92.5	2096.7	19394094.0	0.823E+05
Wearplate	305.0	26.0	198.0	79.3	1570139.8	0.424E+04
Web	20.0	369.0	395.5	73.8	2918790.2	0.623E+05
BasePlate	305.0	30.0	595.0	91.5	5444250.0	0.202E+06
Totals	...	...	...	2341.3	29327272.0	0.351E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 29327272 / 2341.259$$

$$= 125.263 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 148700.719$$

$$= 30263.8184 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 123.7279 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 2134.0 - 1524.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 873.662 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 26440.3848 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 94.4183 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt[3]{(3 \cdot (148701 + 1130) \cdot 305.0 / (4 \cdot 2780.0 \cdot 1767.415))}$$

$$= 26.411 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1524.0 + 185.0 + 26.0)$$

$$= 3005.108 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 3005.1079 / (6 - 1)$$

$$= 601.022 \text{ mm}$$



Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot Bpwid / 2 \\ &= 601.0215 \cdot 305.0 / 2 \\ &= 916.558 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth) \\ &= 149830.562 / (2780.0 \cdot 305.0) \\ &= 17.671 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 916.6 \cdot 17.67 \\ &= 16196.295 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= Rib \text{ Area} + Web \text{ Area} \\ &= 52.0 + 60.102 \\ &= 112.102 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / Ar \\ &= 16196.3 / 112.1022 \\ &= 144.478 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	280.0	...	56.0	...	0.366E+04

Rib dimension [D]:

$$\begin{aligned} &= Saddle \text{ Width} - Web \text{ Thickness} \\ &= 280.0 - 20.0 \\ &= 260.000 \text{ mm} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0.0 / 112.102 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= Saddle \text{ Width} / 2 \\ &= 280.0 / 2 \\ &= 140.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) } \\ &= \sqrt{ (3658.7 / 112.102) } \\ &= 57.129 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned} &= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) - bpthk \\ &= 2134.0 - \cos(120.0 / 2)(1524.0 + 185.0 + 26.0) - 30.0 \\ &= 1236.500 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned} &= 2 \cdot \sin(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) \\ &= 2 \cdot \sin(120.0 / 2)(1524.0 + 185.0 + 26.0) \\ &= 3005.108 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned} &= Baseplate \text{ length} - clearance \\ &= 2780.0 - 50.8 \\ &= 2729.200 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:



$$= (L_w - B_d) / 2$$

$$= (3005.108 - 2729.2) / 2$$

$$= 137.954 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{1236.5^2 + 137.954^2}$$

$$= 1244.172 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 2038900}{2651.1}}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1244.172 / 57.129$$

$$= 21.778$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 755.0 / (2 \cdot 2780.0) \cdot 601.022 \cdot 1244.17 / 2$$

$$= 50.772 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($21.7784 < 123.2106$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (21.78)^2 / (2 \cdot 123.21^2)) 2651 /$$

$$(5/3 + 3 \cdot (21.78) / (8 \cdot 123.21) - (21.78^3) / (8 \cdot 123.21^3))$$

$$= 1507 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 144.48 / 1506.53 + (50.77 \cdot 140.0 / 36586676) / 1767.42$$

$$= 0.107$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	260.0	0.0	52.0	0.0	0.366E+04
Web	601.0	20.0	0.0	120.2	0.0	40.1
Totals	...	...	...	172.2	...	0.370E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 172.204$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 280.0 / 2$$

$$= 140.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (Pitch/2)^2} - B_{pthk}$$

$$= 2134.0 - \sqrt{(1735.0 + 26.0)^2 - (601.022/2)^2} - 30.0$$

$$= 395.223 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{3697.4 / 172.204}$$

$$= 46.337 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 395.223 / 46.337$$

$$= 8.529$$

Unit Force [Force,u]:



$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 755.025 / (2 \cdot 2780.0)$$

$$= 0.136 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.136 \cdot 601.022 \cdot 395.223$$

$$= 32.257 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 32.257 / 264100.125$$

$$= 12.214 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.5294 < 123.2106$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.53)^2 / (2 \cdot 123.21^2)) 2651 / (5/3 + 3 \cdot (8.53) / (8 \cdot 123.21) - (8.53)^3 / (8 \cdot 123.21^3))$$

$$= 1563 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 188.19 / 1562.56 + (32.26 \cdot 140.0 / 3697.399) / 1767.42$$

$$= 0.127$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	75.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.0075 cm ²
Saddle Load QO (Weight)	QO	148191.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	271.1 kgf
Maximum Transverse Force	Ft	711.7 kgf
Maximum Longitudinal Force	Fl	755.0 kgf
Saddle Bolted to Steel Foundation	No	No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 755.0 / (6.008 \cdot 8)$$

$$= 15.7 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 711.7 / (6.008 \cdot 8)$$

$$= 14.8 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 755.03 / (1052.29 \cdot 8.0)$$

$$= 0.0897 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 2134.0 \cdot 711.68 + 0.0$$

$$= 1518.73 \text{ kgf-m.}$$



Eccentricity (e):
= Rmom / QO
= 1518.73/148191.66
= 10.25 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.0897, 0.0]
= 0.0897 cm²

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: NozzleA24"

From: 10

Pressure for Reinforcement Calculations	P	184.342	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1647.49	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Hemispherical Head	D	3137.00	mm
Head Finished (Minimum) Thickness	t	94.0000	mm
Head Internal Corrosion Allowance	cas	3.1750	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	800.0000	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		20.5354	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	44.0000	mm
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	156.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	94.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: NozzleA24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	20.535	in.
Actual Thickness Used in Calculation	1.732	in.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + ci + co$$

$$= 0.5 \cdot 527.95 \cdot (\exp(184.342/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0$$

$$= 40.018 + 3.175 + 0.0 = 43.193 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 40.825 + 527.95) / 527.95) - 0.0$$



$$= 187.805 - 0.0 = 187.8047 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.515 mm

Min. required shell thickness, internal p + c : 93.635 mm

Nozzle neck minimum required thickness : 11.515 mm

Minimum nozzle neck thickness 43.993 mm , is greater than required thickness 11.515 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 702.8,	Allowable	: 1306.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2688.5 kgf/cm ²	Passed
Occasional	: 553.2,	Allowable	: 1737.0 kgf/cm ²	Passed
Shear	: 97.4,	Allowable	: 914.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1306.0/1647.5, 1]$$

$$= 0.793$$

Thickness of Nozzle at Shell [tn]:

= hub thickness - corrosion allowance

$$= 200.0 - 3.175$$

$$= 196.825 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

= thickness - corrosion allowance

$$= 44.0 - 3.175$$

$$= 40.825 \text{ mm}$$

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance

$$= 3137.0/2 + 3.175$$

$$= 1571.675 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1571.675 \cdot 90.825}, 2 \cdot 263.975)$$

$$= 377.819 \text{ mm}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((90.825 + 0.0)/196.825)^{0.35}, 1)$$

$$= \min(0.763, 1)$$

$$= 0.763$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t)$$

$$= \min(90.825 + 0.0 + 0.763 \cdot \sqrt{263.975 \cdot 196.825}, 500.0 + 90.825)$$

$$= 264.711 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + (A_5 \cdot f_{rp}) / LR$$

$$= 90.825 + ((0.0 \cdot 1.0)/377.819)$$

$$= 90.825 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:



$$\begin{aligned} &= t \cdot LR \\ &= 90.825 \cdot 377.819 \\ &= 343.154 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2B \\ &= 474.004 + 44.161 \\ &= 518.165 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg41^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 343.154 + 0.793(518.165 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 754.415 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 196.825 / \ln[1 + 196.825/263.975] \\ &= 353.297 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 90.825 / \ln[1 + 90.825/1571.675] \\ &= 1616.664 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1647.49 \cdot 1.0 \\ &= 2471.2 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 184.342 \cdot 353.297 \cdot 264.711 \\ &= 172399.5 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs \cdot (LR + tn)/2 \\ &= 184.342 \cdot 1616.664 \cdot (377.819 + 196.825)/2 \\ &= 856274.2 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot Rxs \cdot Rnc)/2 \\ &= (184.342 \cdot 1616.664 \cdot 306.677)/2 \\ &= 456977.2 \text{ kgf} \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (856274 + 456977.2 + 172399.5)/184.342 \\ &= 8059.2 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/(2 \cdot teff)) \\ &= 2471.235 / (2 \cdot 8059.216/754.415 - 1616.664/(2 \cdot 90.825)) \\ &= 198.2 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S[t/Rxs] \\ &= 2 \cdot 1647.49[90.825/1616.664] \\ &= 185.1 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(198.245, 185.114) \\ &= 185.114 \text{ kgf/cm}^2 \end{aligned}$$



Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (172399.5 + 856274 + 456977.156) / 754.415 \\ &= 1969.279 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\ &= 184.342 \cdot 1616.664 / (2 \cdot 90.825) \\ &= 1640.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1969.279 - 1640.624, 1640.624) \\ &= 2297.9 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2471.24	kgf/cm ²
Local Primary Membrane Stress	PL	2297.93	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	185.11	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (306.677 + 196.825) / 306.677 \\ &= 1.642 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{((R_{nc} + t_n)^2 + (R_n + t_n)^2) / 2} \\ &= \pi/2 \cdot \sqrt{((306.677 + 196.825)^2 + (263.975 + 196.825)^2) / 2} \\ &= 758.103 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 196.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(456977 \cdot 1.64, 1.5 \cdot 1306.0(597.037 + 0.0), \pi/4 \cdot 184.3 \cdot 263.98^2 \cdot 1.64^2) \\ &= 271944.219 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T)) \\ &= 271944.219 / (758.103(0.49 \cdot 7.071 + 0.6 \cdot 90.825 + 0.49 \cdot 0.0)) \\ &= 618.907 < \text{ or } = \text{ to } 1647.490 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle farther away from the intersecting weld.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	16 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	20 °C
Temperature reduction per Graph 3.13	:	4 °C
The computed stress ratio is	:	0.917

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (83.316 \cdot 1.0) / (94.0 - 3.175 - 0.0) \end{aligned}$$



= 0.917

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 16 °C

The Drop for this Nozzle is : 416.0985 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1024.3162 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: NozzleA24" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.1750	mm
Vessel Diameter	Dv	3137.000	mm
Vessel Thickness	Tv	94.000	mm
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1647.49	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.1750	mm
Nozzle Diameter	Dn	521.600	mm
Nozzle Thickness	Tn	200.000	mm
Nozzle Material		SA-765 I	
Nozzle UNS Number		K03046	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1306.00	kgf/cm ²
Design Internal Pressure	Dp	184.342	kgf/cm ²
Include Pressure Thrust		No	

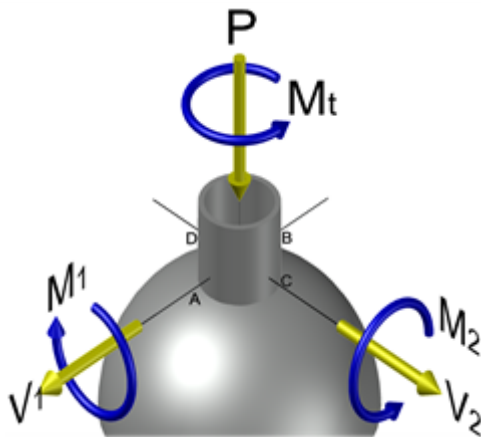
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-7645.2	kgf
Longitudinal Shear (SUS)	(V1) V1	6116.2	kgf
Circumferential Shear (SUS)	(Vc) V2	6116.2	kgf
Circumferential Moment (SUS)	(Mc) M1	11009.2	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	11009.2	kgf-m.
Torsional Moment (SUS)	Mt	12844.0	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 921.6 + 2 \cdot 1.65 \cdot \sqrt{(1617.088 (94.0 - 3.175))}$$

$$= 2186.288 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-7645.2	kgf
Circumferential Shear	(VC)	V2	6116.2	kgf
Longitudinal Shear	(VL)	V1	6116.2	kgf
Circumferential Moment	(MC)	M1	11009.2	kgf-m.
Longitudinal Moment	(ML)	M2	11009.2	kgf-m.
Torsional Moment		MT	12844.0	kgf-m.

Dimensionless Param: U = 1.20 TAU = 5.00 (1.84) RHO = 0.46

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03363	(A,B,C,D)
$M(x) / P$	SP 1	0.03373	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03656	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.05460	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03656	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.05460	(A,B,C,D)
$N(y) * T / P$	SP 1	0.05218	(A,B,C,D)
$M(y) / P$	SP 1	0.00959	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.05115	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01706	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.05115	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01706	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Rad. Bend.	P	18.8	-18.8	18.8	-18.8	18.8	-18.8	18.8	-18.8
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-12.7	-12.7	12.7	12.7
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-114.1	114.1	114.1	-114.1
Rad. Memb.	ML	-12.7	-12.7	12.7	12.7	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-114.1	114.1	114.1	-114.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-104.9	85.7	148.7	-117.0	-104.9	85.7	148.7	-117.0

Tang. Memb. P	4.8	4.8	4.8	4.8	4.8	4.8	4.8	4.8
Tang. Bend. P	5.3	-5.3	5.3	-5.3	5.3	-5.3	5.3	-5.3
Tang. Memb. MC	0.0	0.0	0.0	0.0	-17.8	-17.8	17.8	17.8
Tang. Bend. MC	0.0	0.0	0.0	0.0	-35.6	35.6	35.6	-35.6
Tang. Memb. ML	-17.8	-17.8	17.8	17.8	0.0	0.0	0.0	0.0
Tang. Bend. ML	-35.6	35.6	35.6	-35.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-43.3	17.3	63.6	-18.3	-43.3	17.3	63.6	-18.3
Shear VC	4.7	4.7	-4.7	-4.7	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-4.7	-4.7	4.7	4.7
Shear MT	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6
Tot. Shear	15.3	15.3	5.9	5.9	5.9	5.9	15.3	15.3
Str. Int.	108.5	89.0	149.1	117.3	105.5	86.2	151.3	119.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6
Rad. Pl (SUS)		-9.6	-9.6	15.8	15.8	-9.6	-9.6	15.8	15.8
Rad. Q (SUS)		-95.3	95.3	132.8	-132.8	-95.3	95.3	132.8	-132.8
Long. Pm (SUS)		1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6
Long. Pl (SUS)		-13.0	-13.0	22.7	22.7	-13.0	-13.0	22.7	22.7
Long. Q (SUS)		-30.3	30.3	41.0	-41.0	-30.3	30.3	41.0	-41.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.7	4.7	-4.7	-4.7	-4.7	-4.7	4.7	4.7
Shear Q (SUS)		10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6
Pm (SUS)		1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6
Pm+Pl (SUS)		1634.3	1634.3	1665.6	1665.6	1634.3	1634.3	1665.6	1665.6
Pm+Pl+Q (Total)		1600.9	1729.6	1789.7	1622.7	1597.9	1726.8	1792.0	1624.6

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1640.62	1647.49	Passed
Pm+Pl (SUS)	1665.63	2471.23	Passed
Pm+Pl+Q (TOTAL)	1791.96	4947.71	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: NozzleB24"

From: 10

Pressure for Reinforcement Calculations	P	184.342	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1647.49	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Hemispherical Head	D	3137.00	mm
Head Finished (Minimum) Thickness	t	94.0000	mm
Head Internal Corrosion Allowance	cas	3.1750	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	800.0000	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		20.5354	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	44.0000	mm
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	156.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	94.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: NozzleB24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	20.535	in.
Actual Thickness Used in Calculation	1.732	in.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + ci + co$$

$$= 0.5 \cdot 527.95 \cdot (\exp(184.342/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0$$

$$= 40.018 + 3.175 + 0.0 = 43.193 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 40.825 + 527.95) / 527.95) - 0.0$$



$$= 187.805 - 0.0 = 187.8047 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.515 mm

Min. required shell thickness, internal p + c : 93.635 mm

Nozzle neck minimum required thickness : 11.515 mm

Minimum nozzle neck thickness 43.993 mm , is greater than required thickness 11.515 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 702.8,	Allowable	: 1306.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2688.5 kgf/cm ²	Passed
Occasional	: 553.2,	Allowable	: 1737.0 kgf/cm ²	Passed
Shear	: 97.4,	Allowable	: 914.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1306.0/1647.5, 1] \\ &= 0.793 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 200.0 - 3.175 \\ &= 196.825 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 44.0 - 3.175 \\ &= 40.825 \text{ mm} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3137.0/2 + 3.175 \\ &= 1571.675 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{1571.675 \cdot 90.825}, 2 \cdot 263.975) \\ &= 377.819 \text{ mm} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((90.825 + 0.0)/196.825)^{0.35}, 1) \\ &= \min(0.763, 1) \\ &= 0.763 \end{aligned}$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t) \\ &= \min(90.825 + 0.0 + 0.763 \cdot \sqrt{263.975 \cdot 196.825}, 500.0 + 90.825) \\ &= 264.711 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + (A_5 \cdot f_{rp}) / LR \\ &= 90.825 + ((0.0 \cdot 1.0) / 377.819) \\ &= 90.825 \text{ mm} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:



$$\begin{aligned} &= t \cdot LR \\ &= 90.825 \cdot 377.819 \\ &= 343.154 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2B \\ &= 474.004 + 44.161 \\ &= 518.165 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg41^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 343.154 + 0.793(518.165 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 754.415 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 196.825 / \ln[1 + 196.825/263.975] \\ &= 353.297 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 90.825 / \ln[1 + 90.825/1571.675] \\ &= 1616.664 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1647.49 \cdot 1.0 \\ &= 2471.2 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 184.342 \cdot 353.297 \cdot 264.711 \\ &= 172399.5 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs \cdot (LR + tn)/2 \\ &= 184.342 \cdot 1616.664 \cdot (377.819 + 196.825)/2 \\ &= 856274.2 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot Rxs \cdot Rnc)/2 \\ &= (184.342 \cdot 1616.664 \cdot 306.677)/2 \\ &= 456977.2 \text{ kgf} \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (856274 + 456977.2 + 172399.5)/184.342 \\ &= 8059.2 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/(2 \cdot teff)) \\ &= 2471.235 / (2 \cdot 8059.216/754.415 - 1616.664/(2 \cdot 90.825)) \\ &= 198.2 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S[t/Rxs] \\ &= 2 \cdot 1647.49[90.825/1616.664] \\ &= 185.1 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(198.245, 185.114) \\ &= 185.114 \text{ kgf/cm}^2 \end{aligned}$$



Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (172399.5 + 856274 + 456977.156) / 754.415 \\ &= 1969.279 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\ &= 184.342 \cdot 1616.664 / (2 \cdot 90.825) \\ &= 1640.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1969.279 - 1640.624, 1640.624) \\ &= 2297.9 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2471.24	kgf/cm ²
Local Primary Membrane Stress	PL	2297.93	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	185.11	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (306.677 + 196.825) / 306.677 \\ &= 1.642 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{((R_{nc} + t_n)^2 + (R_n + t_n)^2) / 2} \\ &= \pi/2 \cdot \sqrt{((306.677 + 196.825)^2 + (263.975 + 196.825)^2) / 2} \\ &= 758.103 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 196.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(456977 \cdot 1.64, 1.5 \cdot 1306.0(597.037 + 0.0), \pi/4 \cdot 184.3 \cdot 263.98^2 \cdot 1.64^2) \\ &= 271944.219 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T)) \\ &= 271944.219 / (758.103(0.49 \cdot 7.071 + 0.6 \cdot 90.825 + 0.49 \cdot 0.0)) \\ &= 618.907 < \text{ or } = \text{ to } 1647.490 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle farther away from the intersecting weld.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	16 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	20 °C
Temperature reduction per Graph 3.13	:	4 °C
The computed stress ratio is	:	0.917

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (83.316 \cdot 1.0) / (94.0 - 3.175 - 0.0) \end{aligned}$$



= 0.917

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 16 °C

The Drop for this Nozzle is : 416.0985 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1024.3162 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: NozzleB24" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.1750	mm
Vessel Diameter	Dv	3137.000	mm
Vessel Thickness	Tv	94.000	mm
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1647.49	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.1750	mm
Nozzle Diameter	Dn	521.600	mm
Nozzle Thickness	Tn	200.000	mm
Nozzle Material		SA-765 I	
Nozzle UNS Number		K03046	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1306.00	kgf/cm ²
Design Internal Pressure	Dp	184.342	kgf/cm ²
Include Pressure Thrust		No	

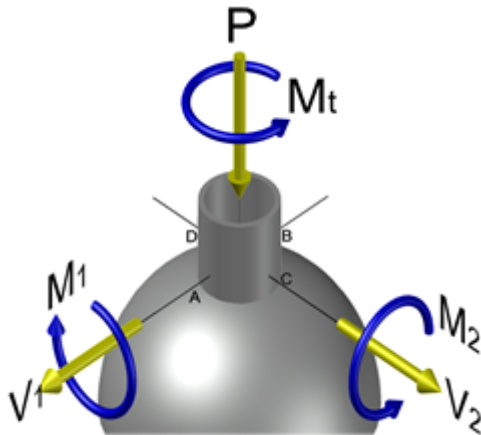
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-7645.2	kgf
Longitudinal Shear (SUS)	(V1) V1	6116.2	kgf
Circumferential Shear (SUS)	(Vc) V2	6116.2	kgf
Circumferential Moment (SUS)	(Mc) M1	11009.1	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	11009.1	kgf-m.
Torsional Moment (SUS)	Mt	12844.0	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 921.6 + 2 \cdot 1.65 \cdot \sqrt{(1617.088 (94.0 - 3.175))}$$

$$= 2186.288 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-7645.2	kgf
Circumferential Shear	(VC) V2	6116.2	kgf
Longitudinal Shear	(VL) V1	6116.2	kgf
Circumferential Moment	(MC) M1	11009.1	kgf-m.
Longitudinal Moment	(ML) M2	11009.1	kgf-m.
Torsional Moment	MT	12844.0	kgf-m.

Dimensionless Param: U = 1.20 TAU = 5.00 (1.84) RHO = 0.46

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03363	(A,B,C,D)
$M(x) / P$	SP 1	0.03373	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03656	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.05460	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03656	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.05460	(A,B,C,D)
$N(y) * T / P$	SP 1	0.05218	(A,B,C,D)
$M(y) / P$	SP 1	0.00959	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.05115	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01706	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.05115	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01706	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Rad. Bend.	P	18.8	-18.8	18.8	-18.8	18.8	-18.8	18.8	-18.8
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-12.7	-12.7	12.7	12.7
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-114.1	114.1	114.1	-114.1
Rad. Memb.	ML	-12.7	-12.7	12.7	12.7	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-114.1	114.1	114.1	-114.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-104.9	85.7	148.7	-117.0	-104.9	85.7	148.7	-117.0

Tang. Memb. P	4.8	4.8	4.8	4.8	4.8	4.8	4.8	4.8
Tang. Bend. P	5.3	-5.3	5.3	-5.3	5.3	-5.3	5.3	-5.3
Tang. Memb. MC	0.0	0.0	0.0	0.0	-17.8	-17.8	17.8	17.8
Tang. Bend. MC	0.0	0.0	0.0	0.0	-35.6	35.6	35.6	-35.6
Tang. Memb. ML	-17.8	-17.8	17.8	17.8	0.0	0.0	0.0	0.0
Tang. Bend. ML	-35.6	35.6	35.6	-35.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-43.3	17.3	63.6	-18.3	-43.3	17.3	63.6	-18.3
Shear VC	4.7	4.7	-4.7	-4.7	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-4.7	-4.7	4.7	4.7
Shear MT	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6
Tot. Shear	15.3	15.3	5.9	5.9	5.9	5.9	15.3	15.3
Str. Int.	108.5	89.0	149.1	117.3	105.5	86.2	151.3	119.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6
Rad. Pl (SUS)		-9.6	-9.6	15.8	15.8	-9.6	-9.6	15.8	15.8
Rad. Q (SUS)		-95.3	95.3	132.8	-132.8	-95.3	95.3	132.8	-132.8
Long. Pm (SUS)		1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6
Long. Pl (SUS)		-13.0	-13.0	22.7	22.7	-13.0	-13.0	22.7	22.7
Long. Q (SUS)		-30.3	30.3	41.0	-41.0	-30.3	30.3	41.0	-41.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.7	4.7	-4.7	-4.7	-4.7	-4.7	4.7	4.7
Shear Q (SUS)		10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6
Pm (SUS)		1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6	1640.6
Pm+Pl (SUS)		1634.3	1634.3	1665.6	1665.6	1634.3	1634.3	1665.6	1665.6
Pm+Pl+Q (Total)		1600.9	1729.6	1789.7	1622.7	1597.9	1726.8	1792.0	1624.6

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1640.62	1647.49	Passed
Pm+Pl (SUS)	1665.63	2471.23	Passed
Pm+Pl+Q (TOTAL)	1791.95	4947.71	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: BOOT

From: 20

Pressure for Reinforcement Calculations	P	184.387	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1647.49	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	3048.00	mm
Shell Finished (Minimum) Thickness	t	185.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		100.00	cm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	44.0000	mm
Hub Height of Integral Nozzle	h	200.0000	mm
Height of Beveled Transition	L'	156.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	610.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	184.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: BOOT

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	20.000	in.
Actual Thickness Used in Calculation	1.732	in.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + ci + co$$

$$= 0.5 \cdot 514.35 \cdot (\exp(184.387/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0$$

$$= 38.9973 + 3.175 + 0.0 = 42.1723 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - Pliq$$



$$= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 40.825 + 514.35) / 514.35) - 0.0$$
$$= 192.422 - 0.0 = 192.4218 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.515 mm

Min. required shell thickness, internal p + c : 184.228 mm

Nozzle neck minimum required thickness : 11.515 mm

Minimum nozzle neck thickness 43.993 mm , is greater than required thickness 11.515 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1306.0/1647.5, 1]$$

$$= 0.793$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 200.0 - 3.175$$

$$= 196.825 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 44.0 - 3.175$$

$$= 40.825 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 3054.35/181.825$$

$$= 16.798 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 3048.0/2 + 3.175$$

$$= 1527.175 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1527.175 \cdot 181.825}, 2 \cdot 257.175)$$

$$= 514.350 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 181.825, 0.0] + \sqrt{257.175 \cdot 196.825}$$

$$= 224.985 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 610.0$$

$$= 610.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(181.825 + 0.0)$$

$$= 1454.600 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[224.985, 610.0, 1454.6] + 181.825$$

$$= 406.810 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 181.825 + ((0.0 \cdot 1.0) / 514.35)$$

$$= 181.825 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$



$$= \min(12, (2 \cdot 257.17 + 196.825) / (\sqrt{(3054.35 + 181.825) \cdot 181.825})) \\ = 0.927$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ = 181.825 \cdot 514.35 \cdot \max((0.927/5)^{0.85}, 1) \\ = 935.217 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B \\ = 751.527 + 46.056 \\ = 797.583 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2 \\ = 0.5 \cdot 10.0^2 \\ = 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ = 935.217 + 0.793(797.583 + 0.0) + 0.5 + 0.0 + 0.0 \\ = 1567.978 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n] \\ = 196.825 / \ln[1 + 196.825/257.175] \\ = 346.315 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ = 181.825 / \ln[1 + 181.825/1527.175] \\ = 1616.384 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E \\ = 1.5 \cdot 1647.49 \cdot 1.0 \\ = 2471.2 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H \\ = 184.387 \cdot 346.315 \cdot 406.81 \\ = 259773.1 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n) \\ = 184.387 \cdot 1616.384 \cdot (514.35 + 196.825) \\ = 2119587.0 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc} \\ = 184.387 \cdot 1616.384 \cdot 257.175 \\ = 766484.8 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P \\ = (2119587 + 766485 + 259773.1) / 184.387 \\ = 17061.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ = 2471.235 / (2 \cdot 17061.1 / 1567.978 - 1616.384 / 181.825) \\ = 192.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{xs}] \\ = 1647.49[181.825/1616.384] \\ = 185.3 \text{ kgf/cm}^2$$



Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(191.983, 185.324) \\ &= 185.324 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (259773.125 + 2119587 + 766485) / 1567.978 \\ &= 2006.310 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_s / t_{eff} \\ &= 184.387 \cdot 1616.384 / 181.825 \\ &= 1639.2 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 2006.31 - 1639.162, 1639.162) \\ &= 2373.5 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2471.24	kgf/cm ²
Local Primary Membrane Stress	PL	2373.46	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	185.32	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (257.175 + 196.825) / 257.175 \\ &= 1.765 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (257.175 + 196.825) \\ &= 713.142 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 196.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(766485 \cdot 1.77, 1.5 \cdot 1306.0(797.583 + 0.0), \pi/4 \cdot 184.4 \cdot 257.17^2 \cdot 1.77^2) \\ &= 298491.438 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L41T + 0.6 \cdot t_w1 + 0.49 \cdot L43T)) \\ &= 298491.438 / (713.142(0.49 \cdot 7.071 + 0.6 \cdot 181.825 + 0.49 \cdot 0.0)) \\ &= 371.855 < \text{ or = to } 1647.490 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

The Drop for this Nozzle is : 69.1942 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 864.1943 mm



Input, Nozzle Desc: L22"

Parent: BOOT

Pressure for Reinforcement Calculations	P	184.401	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-765 I	+IT
Parent Allowable Stress at Temperature	Sv	1306.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	508.00	mm
Shell Finished (Minimum) Thickness	t	44.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-350 LF2	
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	184.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L22"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1



Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 57.15 \cdot (\exp(184.401/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0 \\ &= 4.3334 + 3.175 + 0.0 = 7.5084 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 6.825 + 57.15) / 57.15) - 0.0 \\ &= 279.719 - 0.0 = 279.7187 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.695 mm

Min. required shell thickness, internal p + c : 42.245 mm

Nozzle neck minimum required thickness : 7.695 mm

Minimum nozzle neck thickness 10.008 mm , is greater than required thickness 7.695 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1306.0/1306.0, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 50.0 - 3.175 \\ &= 46.825 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 10.0 - 3.175 \\ &= 6.825 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 514.35/40.825 \\ &= 12.599 \text{ must be less than } 400. \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 508.0/2 + 3.175 \\ &= 257.175 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{257.175 \cdot 40.825}, 2 \cdot 28.575) \\ &= 57.150 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\ &= \min[1.5 \cdot 40.825, 0.0] + \sqrt{28.575 \cdot 46.825} \\ &= 36.579 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(40.825 + 0.0) \\ &= 326.600 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\ &= \min[36.579, 300.0, 326.6] + 40.825 \\ &= 77.404 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 40.825 + ((0.0 \cdot 1.0) / 57.15) \\ &= 40.825 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 28.57 + 46.825) / (\sqrt{(514.35 + 40.825) \cdot 40.825})) \\ &= 0.691 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 40.825 \cdot 57.15 \cdot \max((0.691/5)^{0.85}, 1) \\ &= 23.331 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= tn \cdot LH \\ &= 46.825 \cdot 77.404 \\ &= 36.244 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg41^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 23.331 + 1.0(36.244 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 60.076 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 46.825 / \ln[1 + 46.825/28.575] \\ &= 48.260 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 40.825 / \ln[1 + 40.825/257.175] \\ &= 277.086 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1306.0 \cdot 1.0 \\ &= 1959.0 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 184.401 \cdot 48.26 \cdot 77.404 \\ &= 6888.2 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs (LR + tn) \\ &= 184.401 \cdot 277.086 (57.15 + 46.825) \\ &= 53125.8 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 184.401 \cdot 277.086 \cdot 28.575 \\ &= 14600.3 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (53125.8 + 14600.3 + 6888.2) / 184.401 \\ &= 404.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$



$$= 1959.0 / (2 \cdot 404.633 / 60.076 - 277.086 / 40.825)$$
$$= 293.1 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$
$$= 1306.0 [40.825 / 277.086]$$
$$= 192.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(293.108, 192.422)$$
$$= 192.422 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$
$$= (6888.238 + 53125.844 + 14600.345) / 60.076$$
$$= 1242.004 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff$$
$$= 184.401 \cdot 277.086 / 40.825$$
$$= 1251.6 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 1242.004 - 1251.559, 1251.559)$$
$$= 1251.6 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	1959.00	kgf/cm ²
Local Primary Membrane Stress	PL	1251.56	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	192.42	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (28.575 + 46.825) / 28.575$$
$$= 2.639 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$
$$= \pi/2 \cdot (28.575 + 46.825)$$
$$= 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 46.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$
$$= \min(14600 \cdot 2.64, 1.5 \cdot 1306.0(36.244 + 0.0), \pi/4 \cdot 184.4 \cdot 28.57^2 \cdot 2.64^2)$$
$$= 8233.688 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$
$$= 8233.688 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 40.825 + 0.49 \cdot 0.0))$$
$$= 248.639 < \text{ or } = \text{ to } 1306.000 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	1 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	7 °C



Temperature reduction per Graph 3.13 : 6 °C
The computed stress ratio is : 0.882

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (36.014 • 1.0)/(44.0 - 3.175 - 0.0)
= 0.882

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 1 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 11.4493 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 355.4493 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: L42"

Parent: BOOT

Pressure for Reinforcement Calculations	P	184.401	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-765 I	+IT
Parent Allowable Stress at Temperature	Sv	1306.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	508.00	mm
Shell Finished (Minimum) Thickness	t	44.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		250.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-350 LF2	
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	184.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L42"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1



Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 57.15 \cdot (\exp(184.401/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0 \\ &= 4.3334 + 3.175 + 0.0 = 7.5084 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 6.825 + 57.15) / 57.15) - 0.0 \\ &= 279.719 - 0.0 = 279.7187 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.695 mm

Min. required shell thickness, internal p + c : 42.245 mm

Nozzle neck minimum required thickness : 7.695 mm

Minimum nozzle neck thickness 10.008 mm , is greater than required thickness 7.695 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1306.0/1306.0, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 50.0 - 3.175 \\ &= 46.825 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 10.0 - 3.175 \\ &= 6.825 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 514.35/40.825 \\ &= 12.599 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 508.0/2 + 3.175 \\ &= 257.175 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{257.175 \cdot 40.825}, 2 \cdot 28.575) \\ &= 57.150 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\ &= \min[1.5 \cdot 40.825, 0.0] + \sqrt{28.575 \cdot 46.825} \\ &= 36.579 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(40.825 + 0.0) \\ &= 326.600 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\ &= \min[36.579, 300.0, 326.6] + 40.825 \\ &= 77.404 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 40.825 + ((0.0 \cdot 1.0) / 57.15) \\ &= 40.825 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 28.57 + 46.825) / (\sqrt{(514.35 + 40.825) \cdot 40.825})) \\ &= 0.691 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 40.825 \cdot 57.15 \cdot \max((0.691/5)^{0.85}, 1) \\ &= 23.331 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= tn \cdot LH \\ &= 46.825 \cdot 77.404 \\ &= 36.244 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg41^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 23.331 + 1.0(36.244 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 60.076 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 46.825 / \ln[1 + 46.825/28.575] \\ &= 48.260 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 40.825 / \ln[1 + 40.825/257.175] \\ &= 277.086 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1306.0 \cdot 1.0 \\ &= 1959.0 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 184.401 \cdot 48.26 \cdot 77.404 \\ &= 6888.2 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs (LR + tn) \\ &= 184.401 \cdot 277.086 (57.15 + 46.825) \\ &= 53125.8 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 184.401 \cdot 277.086 \cdot 28.575 \\ &= 14600.3 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (53125.8 + 14600.3 + 6888.2) / 184.401 \\ &= 404.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$



$$= 1959.0 / (2 \cdot 404.633 / 60.076 - 277.086 / 40.825)$$
$$= 293.1 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$
$$= 1306.0 [40.825 / 277.086]$$
$$= 192.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(293.108, 192.422)$$
$$= 192.422 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$
$$= (6888.238 + 53125.844 + 14600.345) / 60.076$$
$$= 1242.004 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff$$
$$= 184.401 \cdot 277.086 / 40.825$$
$$= 1251.6 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 1242.004 - 1251.559, 1251.559)$$
$$= 1251.6 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	1959.00	kgf/cm ²
Local Primary Membrane Stress	PL	1251.56	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	192.42	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (28.575 + 46.825) / 28.575$$
$$= 2.639 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$
$$= \pi / 2 \cdot (28.575 + 46.825)$$
$$= 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 46.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$
$$= \min(14600 \cdot 2.64, 1.5 \cdot 1306.0 (36.244 + 0.0), \pi / 4 \cdot 184.4 \cdot 28.57^2 \cdot 2.64^2)$$
$$= 8233.688 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt (0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$
$$= 8233.688 / (118.438 (0.49 \cdot 7.071 + 0.6 \cdot 40.825 + 0.49 \cdot 0.0))$$
$$= 248.639 < \text{ or } = \text{ to } 1306.000 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	1 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	7 °C



Temperature reduction per Graph 3.13 : 6 °C
The computed stress ratio is : 0.882

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (36.014 • 1.0)/(44.0 - 3.175 - 0.0)
= 0.882

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 1 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 11.4493 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 355.4493 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: L62"

Parent: BOOT

Pressure for Reinforcement Calculations	P	184.401	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-765 I	+IT
Parent Allowable Stress at Temperature	Sv	1306.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	508.00	mm
Shell Finished (Minimum) Thickness	t	44.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		220.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-350 LF2	
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	184.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L62"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1



Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 57.15 \cdot (\exp(184.401/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0 \\ &= 4.3334 + 3.175 + 0.0 = 7.5084 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 6.825 + 57.15) / 57.15) - 0.0 \\ &= 279.719 - 0.0 = 279.7187 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.695 mm

Min. required shell thickness, internal p + c : 42.245 mm

Nozzle neck minimum required thickness : 7.695 mm

Minimum nozzle neck thickness 10.008 mm , is greater than required thickness 7.695 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1306.0/1306.0, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 50.0 - 3.175 \\ &= 46.825 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 10.0 - 3.175 \\ &= 6.825 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 514.35/40.825 \\ &= 12.599 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 508.0/2 + 3.175 \\ &= 257.175 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{257.175 \cdot 40.825}, 2 \cdot 28.575) \\ &= 57.150 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\ &= \min[1.5 \cdot 40.825, 0.0] + \sqrt{28.575 \cdot 46.825} \\ &= 36.579 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(40.825 + 0.0) \\ &= 326.600 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\ &= \min[36.579, 300.0, 326.6] + 40.825 \\ &= 77.404 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 40.825 + ((0.0 \cdot 1.0) / 57.15) \\ &= 40.825 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 28.57 + 46.825) / (\sqrt{(514.35 + 40.825) \cdot 40.825})) \\ &= 0.691 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 40.825 \cdot 57.15 \cdot \max((0.691/5)^{0.85}, 1) \\ &= 23.331 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= tn \cdot LH \\ &= 46.825 \cdot 77.404 \\ &= 36.244 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg41^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 23.331 + 1.0(36.244 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 60.076 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 46.825 / \ln[1 + 46.825/28.575] \\ &= 48.260 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 40.825 / \ln[1 + 40.825/257.175] \\ &= 277.086 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1306.0 \cdot 1.0 \\ &= 1959.0 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 184.401 \cdot 48.26 \cdot 77.404 \\ &= 6888.2 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs (LR + tn) \\ &= 184.401 \cdot 277.086 (57.15 + 46.825) \\ &= 53125.8 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 184.401 \cdot 277.086 \cdot 28.575 \\ &= 14600.3 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (53125.8 + 14600.3 + 6888.2) / 184.401 \\ &= 404.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$



$$= 1959.0 / (2 \cdot 404.633 / 60.076 - 277.086 / 40.825)$$
$$= 293.1 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$
$$= 1306.0 [40.825 / 277.086]$$
$$= 192.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(293.108, 192.422)$$
$$= 192.422 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$
$$= (6888.238 + 53125.844 + 14600.345) / 60.076$$
$$= 1242.004 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / t_{eff}$$
$$= 184.401 \cdot 277.086 / 40.825$$
$$= 1251.6 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 1242.004 - 1251.559, 1251.559)$$
$$= 1251.6 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	1959.00	kgf/cm ²
Local Primary Membrane Stress	PL	1251.56	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	192.42	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (28.575 + 46.825) / 28.575$$
$$= 2.639 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$
$$= \pi / 2 \cdot (28.575 + 46.825)$$
$$= 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 46.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$
$$= \min(14600 \cdot 2.64, 1.5 \cdot 1306.0 (36.244 + 0.0), \pi / 4 \cdot 184.4 \cdot 28.57^2 \cdot 2.64^2)$$
$$= 8233.688 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt (0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$
$$= 8233.688 / (118.438 (0.49 \cdot 7.071 + 0.6 \cdot 40.825 + 0.49 \cdot 0.0))$$
$$= 248.639 < \text{ or } = \text{ to } 1306.000 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	1 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	7 °C



Temperature reduction per Graph 3.13 : 6 °C
The computed stress ratio is : 0.882

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (36.014 • 1.0)/(44.0 - 3.175 - 0.0)
= 0.882

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 1 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 11.4493 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 355.4493 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: L82"

Parent: BOOT

Pressure for Reinforcement Calculations	P	184.401	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-765 I	+IT
Parent Allowable Stress at Temperature	Sv	1306.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	508.00	mm
Shell Finished (Minimum) Thickness	t	44.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		200.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-350 LF2	
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	184.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L82"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.394	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1



Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 57.15 \cdot (\exp(184.401/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0 \\ &= 4.3334 + 3.175 + 0.0 = 7.5084 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 6.825 + 57.15) / 57.15) - 0.0 \\ &= 279.719 - 0.0 = 279.7187 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.695 mm

Min. required shell thickness, internal p + c : 42.245 mm

Nozzle neck minimum required thickness : 7.695 mm

Minimum nozzle neck thickness 10.008 mm , is greater than required thickness 7.695 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1306.0/1306.0, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 50.0 - 3.175 \\ &= 46.825 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 10.0 - 3.175 \\ &= 6.825 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 514.35/40.825 \\ &= 12.599 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 508.0/2 + 3.175 \\ &= 257.175 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{257.175 \cdot 40.825}, 2 \cdot 28.575) \\ &= 57.150 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\ &= \min[1.5 \cdot 40.825, 0.0] + \sqrt{28.575 \cdot 46.825} \\ &= 36.579 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(40.825 + 0.0) \\ &= 326.600 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\ &= \min[36.579, 300.0, 326.6] + 40.825 \\ &= 77.404 \text{ mm} \end{aligned}$$



Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 40.825 + ((0.0 \cdot 1.0) / 57.15) \\ &= 40.825 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 28.57 + 46.825) / (\sqrt{(514.35 + 40.825) \cdot 40.825})) \\ &= 0.691 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 40.825 \cdot 57.15 \cdot \max((0.691/5)^{0.85}, 1) \\ &= 23.331 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= tn \cdot LH \\ &= 46.825 \cdot 77.404 \\ &= 36.244 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 23.331 + 1.0(36.244 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 60.076 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 46.825 / \ln[1 + 46.825/28.575] \\ &= 48.260 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 40.825 / \ln[1 + 40.825/257.175] \\ &= 277.086 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1306.0 \cdot 1.0 \\ &= 1959.0 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 184.401 \cdot 48.26 \cdot 77.404 \\ &= 6888.2 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs (LR + tn) \\ &= 184.401 \cdot 277.086 (57.15 + 46.825) \\ &= 53125.8 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 184.401 \cdot 277.086 \cdot 28.575 \\ &= 14600.3 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (53125.8 + 14600.3 + 6888.2) / 184.401 \\ &= 404.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$



$$= 1959.0 / (2 \cdot 404.633 / 60.076 - 277.086 / 40.825) \\ = 293.1 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs] \\ = 1306.0 [40.825 / 277.086] \\ = 192.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2) \\ = \min(293.108, 192.422) \\ = 192.422 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT \\ = (6888.238 + 53125.844 + 14600.345) / 60.076 \\ = 1242.004 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff \\ = 184.401 \cdot 277.086 / 40.825 \\ = 1251.6 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ = \max(2 \cdot 1242.004 - 1251.559, 1251.559) \\ = 1251.6 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	1959.00	kgf/cm ²
Local Primary Membrane Stress	PL	1251.56	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	192.42	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc \\ = (28.575 + 46.825) / 28.575 \\ = 2.639 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn) \\ = \pi/2 \cdot (28.575 + 46.825) \\ = 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 46.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ = \min(14600 \cdot 2.64, 1.5 \cdot 1306.0(36.244 + 0.0), \pi/4 \cdot 184.4 \cdot 28.57^2 \cdot 2.64^2) \\ = 8233.688 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ = 8233.688 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 40.825 + 0.49 \cdot 0.0)) \\ = 248.639 < \text{ or } = \text{ to } 1306.000 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	1 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	7 °C



Temperature reduction per Graph 3.13 : 6 °C
The computed stress ratio is : 0.882

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (36.014 • 1.0)/(44.0 - 3.175 - 0.0)
= 0.882

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 1 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 11.4493 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 355.4493 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K9 2"

From: 20

Pressure for Reinforcement Calculations	P	184.342	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1647.49	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	3048.00	mm
Shell Finished (Minimum) Thickness	t	185.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		150.00	cm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	82.3546	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	184.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K9 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.773	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P / (S \cdot E)) - 1) + ci + co$$

$$= 0.5 \cdot 51.384 \cdot (\exp(184.342 / (1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0$$

$$= 3.8949 + 3.175 + 0.0 = 7.0699 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - Pliq$$



$$= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 4.47 + 51.384) / 51.384) - 0.0$$
$$= 209.503 - 0.0 = 209.5031 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.595 mm

Min. required shell thickness, internal p + c : 184.181 mm

Nozzle neck minimum required thickness : 6.595 mm

Minimum nozzle neck thickness 7.645 mm , is greater than required thickness 6.595 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

= min[Sn/S, 1]

= min[1306.0/1647.5, 1]

= 0.793

Thickness of Nozzle at Shell [tn]:

= hub thickness - corrosion allowance

= 90.0 - 3.175

= 86.825 mm

Thickness of Nozzle at Top [tn2]:

= thickness - corrosion allowance

= 7.645 - 3.175

= 4.470 mm

Shell Diameter to Thickness ratio [D/t]:

= Di/t

= 3054.35/181.825

= 16.798 must be less than 400.

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance

= 3048.0/2 + 3.175

= 1527.175 mm

Effective Length of Vessel Wall [LR]:

= min($\sqrt{(Reff \cdot t)}$, $2 \cdot Rn$)

= min($\sqrt{(1527.175 \cdot 181.825)}$, $2 \cdot 25.692$)

= 51.384 mm

Thickness Limit Candidate [LH1]:

= min[1.5t,te] + $\sqrt{(Rn \cdot tn)}$

= min[1.5•181.825, 0.0] + $\sqrt{(25.692 \cdot 86.825)}$

= 47.230 mm

Thickness Limit Candidate [LH2]:

= Lpr1

= 300.0

= 300.000 mm

Thickness Limit Candidate [LH3]:

= 8(t + te)

= 8(181.825 + 0.0)

= 1454.600 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= min[LH1, LH2, LH3] + t for set-in nozzles

= min[47.23, 300.0, 1454.6] + 181.825

= 229.055 mm

Effective Vessel Thickness [teff]:

= t + ((A5 • frp) / LR)

= 181.825 + ((0.0 • 1.0) / 51.384)

= 181.825 mm

Parameter [λ]:

= min(12, (2•Rn + tn) / ($\sqrt{(Di + teff) \cdot teff}$))



$$= \min(12, (2 \cdot 25.69 + 86.825) / (\sqrt{((3054.35 + 181.825) \cdot 181.825)})) \\ = 0.180$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ = 181.825 \cdot 51.384 \cdot \max((0.18/5)^{0.85}, 1) \\ = 93.429 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH \\ = 86.825 \cdot 229.055 \\ = 198.877 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2 \\ = 0.5 \cdot 10.0^2 \\ = 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ = 93.429 + 0.793(198.877 + 0.0) + 0.5 + 0.0 + 0.0 \\ = 251.584 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n] \\ = 86.825 / \ln[1 + 86.825/25.692] \\ = 58.788 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ = 181.825 / \ln[1 + 181.825/1527.175] \\ = 1616.384 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E \\ = 1.5 \cdot 1647.49 \cdot 1.0 \\ = 2471.2 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH \\ = 184.342 \cdot 58.788 \cdot 229.055 \\ = 24822.9 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n) \\ = 184.342 \cdot 1616.384 (51.384 + 86.825) \\ = 411818.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc} \\ = 184.342 \cdot 1616.384 \cdot 25.692 \\ = 76554.0 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P \\ = (411818.1 + 76554.0 + 24822.9) / 184.342 \\ = 2783.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ = 2471.235 / (2 \cdot 2783.931 / 251.584 - 1616.384 / 181.825) \\ = 186.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{xs}] \\ = 1647.49 [181.825 / 1616.384] \\ = 185.3 \text{ kgf/cm}^2$$



Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(186.628, 185.324) \\ &= 185.324 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (24822.871 + 411818.094 + 76554.039) / 251.584 \\ &= 2039.861 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 184.342 \cdot 1616.384 / 181.825 \\ &= 1638.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 2039.861 - 1638.76, 1638.76) \\ &= 2441.0 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2471.24	kgf/cm ²
Local Primary Membrane Stress	PL	2440.96	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	185.32	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (25.692 + 86.825) / 25.692 \\ &= 4.379 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (25.692 + 86.825) \\ &= 176.741 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 86.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(76554 \cdot 4.38, 1.5 \cdot 1306.0(198.877+0.0), \pi/4 \cdot 184.3 \cdot 25.69^2 \cdot 4.38^2) \\ &= 18329.516 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T)) \\ &= 18329.516 / (176.741(0.49 \cdot 7.071 + 0.6 \cdot 181.825 + 0.49 \cdot 0.0)) \\ &= 92.136 < \text{ or } = \text{ to } 1647.490 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: -33 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.041

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (3.597 \cdot 1.0) / (90.0 - 3.175 - 0.0) \\ &= 0.041 \end{aligned}$$



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

The Drop for this Nozzle is : 4.1592 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 489.1593 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K10 2"

From: 20

Pressure for Reinforcement Calculations	P	184.342	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1647.49	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	3048.00	mm
Shell Finished (Minimum) Thickness	t	185.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		200.00	cm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	82.3546	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	184.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K10 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.773	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + ci + co$$

$$= 0.5 \cdot 51.384 \cdot (\exp(184.342/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0$$

$$= 3.8949 + 3.175 + 0.0 = 7.0699 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - Pliq$$



$$= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 4.47 + 51.384) / 51.384) - 0.0$$
$$= 209.503 - 0.0 = 209.5031 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.595 mm

Min. required shell thickness, internal p + c : 184.181 mm

Nozzle neck minimum required thickness : 6.595 mm

Minimum nozzle neck thickness 7.645 mm , is greater than required thickness 6.595 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1306.0/1647.5, 1]$$

$$= 0.793$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 90.0 - 3.175$$

$$= 86.825 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 7.645 - 3.175$$

$$= 4.470 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 3054.35/181.825$$

$$= 16.798 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 3048.0/2 + 3.175$$

$$= 1527.175 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1527.175 \cdot 181.825}, 2 \cdot 25.692)$$

$$= 51.384 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 181.825, 0.0] + \sqrt{25.692 \cdot 86.825}$$

$$= 47.230 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(181.825 + 0.0)$$

$$= 1454.600 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[47.23, 300.0, 1454.6] + 181.825$$

$$= 229.055 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 181.825 + ((0.0 \cdot 1.0) / 51.384)$$

$$= 181.825 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$



$$= \min(12, (2 \cdot 25.69 + 86.825) / (\sqrt{(3054.35 + 181.825) \cdot 181.825})) \\ = 0.180$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ = 181.825 \cdot 51.384 \cdot \max((0.18/5)^{0.85}, 1) \\ = 93.429 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH \\ = 86.825 \cdot 229.055 \\ = 198.877 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2 \\ = 0.5 \cdot 10.0^2 \\ = 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ = 93.429 + 0.793(198.877 + 0.0) + 0.5 + 0.0 + 0.0 \\ = 251.584 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n] \\ = 86.825 / \ln[1 + 86.825/25.692] \\ = 58.788 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ = 181.825 / \ln[1 + 181.825/1527.175] \\ = 1616.384 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E \\ = 1.5 \cdot 1647.49 \cdot 1.0 \\ = 2471.2 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH \\ = 184.342 \cdot 58.788 \cdot 229.055 \\ = 24822.9 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n) \\ = 184.342 \cdot 1616.384 (51.384 + 86.825) \\ = 411818.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc} \\ = 184.342 \cdot 1616.384 \cdot 25.692 \\ = 76554.0 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P \\ = (411818.1 + 76554.0 + 24822.9) / 184.342 \\ = 2783.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ = 2471.235 / (2 \cdot 2783.931 / 251.584 - 1616.384 / 181.825) \\ = 186.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{xs}] \\ = 1647.49 [181.825 / 1616.384] \\ = 185.3 \text{ kgf/cm}^2$$



Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(186.628, 185.324) \\ &= 185.324 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (24822.871 + 411818.094 + 76554.039) / 251.584 \\ &= 2039.861 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 184.342 \cdot 1616.384 / 181.825 \\ &= 1638.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 2039.861 - 1638.76, 1638.76) \\ &= 2441.0 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2471.24	kgf/cm ²
Local Primary Membrane Stress	PL	2440.96	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	185.32	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (25.692 + 86.825) / 25.692 \\ &= 4.379 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (25.692 + 86.825) \\ &= 176.741 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 86.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(76554 \cdot 4.38, 1.5 \cdot 1306.0(198.877+0.0), \pi/4 \cdot 184.3 \cdot 25.69^2 \cdot 4.38^2) \\ &= 18329.516 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T)) \\ &= 18329.516 / (176.741(0.49 \cdot 7.071 + 0.6 \cdot 181.825 + 0.49 \cdot 0.0)) \\ &= 92.136 < \text{ or } = \text{ to } 1647.490 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: -33 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.041

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (3.597 \cdot 1.0) / (90.0 - 3.175 - 0.0) \\ &= 0.041 \end{aligned}$$



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

The Drop for this Nozzle is : 4.1592 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 489.1593 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: L12"

From: 30

Pressure for Reinforcement Calculations	P	184.342	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1647.49	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	3048.00	mm
Shell Finished (Minimum) Thickness	t	185.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		350.00	cm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		1.2126	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	184.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L12"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.213	in.
Actual Thickness Used in Calculation	0.394	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 37.15 \cdot (\exp(184.342/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0 \\ &= 2.8159 + 3.175 + 0.0 = 5.9909 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 6.825 + 37.15) / 37.15) - 0.0 \\ &= 408.69 - 0.0 = 408.69 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

$$\begin{aligned} \text{Min. required nozzle neck thickness, table 4.5.2 + c} &: 6.595 \text{ mm} \\ \text{Min. required shell thickness, internal p + c} &: 184.181 \text{ mm} \end{aligned}$$

$$\text{Nozzle neck minimum required thickness} : 6.595 \text{ mm}$$

Minimum nozzle neck thickness 10.008 mm, is greater than required thickness 6.595 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1306.0/1647.5, 1] \\ &= 0.793 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 90.0 - 3.175 \\ &= 86.825 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 10.0 - 3.175 \\ &= 6.825 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 3054.35/181.825 \\ &= 16.798 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3048.0/2 + 3.175 \\ &= 1527.175 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{(Reff \cdot t)}, 2 \cdot R_n) \\ &= \min(\sqrt{(1527.175 \cdot 181.825)}, 2 \cdot 18.575) \\ &= 37.150 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 181.825, 0.0] + \sqrt{(18.575 \cdot 86.825)} \\ &= 40.159 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.825 + 0.0) \\ &= 1454.600 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$



$$= \min[40.159, 300.0, 1454.6] + 181.825 \\ = 221.984 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR) \\ = 181.825 + ((0.0 \cdot 1.0) / 37.15) \\ = 181.825 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\ = \min(12, (2 \cdot 18.58 + 86.825) / (\sqrt{(3054.35 + 181.825) \cdot 181.825})) \\ = 0.162$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ = 181.825 \cdot 37.15 \cdot \max((0.162/5)^{0.85}, 1) \\ = 67.548 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH \\ = 86.825 \cdot 221.984 \\ = 192.738 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2 \\ = 0.5 \cdot 10.0^2 \\ = 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ = 67.548 + 0.793(192.738 + 0.0) + 0.5 + 0.0 + 0.0 \\ = 220.835 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn] \\ = 86.825 / \ln[1 + 86.825/18.575] \\ = 50.016 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff] \\ = 181.825 / \ln[1 + 181.825/1527.175] \\ = 1616.384 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E \\ = 1.5 \cdot 1647.49 \cdot 1.0 \\ = 2471.2 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH \\ = 184.342 \cdot 50.016 \cdot 221.984 \\ = 20467.0 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn) \\ = 184.342 \cdot 1616.384 \cdot (37.15 + 86.825) \\ = 369404.9 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc \\ = 184.342 \cdot 1616.384 \cdot 18.575 \\ = 55347.4 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P \\ = (369404.9 + 55347.4 + 20467.0) / 184.342 \\ = 2415.2 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff}) \\ &= 2471.235 / (2 \cdot 2415.183/220.835 - 1616.384/181.825) \\ &= 190.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= \text{S}[t/\text{Rxs}] \\ &= 1647.49[181.825/1616.384] \\ &= 185.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(\text{Pmax1}, \text{Pmax2}) \\ &= \min(190.338, 185.324) \\ &= 185.324 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (\text{fN} + \text{fS} + \text{fY}) / \text{AT} \\ &= (20467.037 + 369404.875 + 55347.414) / 220.835 \\ &= 2016.072 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot \text{Rxs} / \text{teff} \\ &= 184.342 \cdot 1616.384/181.825 \\ &= 1638.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 2016.072 - 1638.76, 1638.76) \\ &= 2393.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2471.24	kgf/cm ²
Local Primary Membrane Stress	PL	2393.38	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	185.32	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (\text{Rnc} + \text{tn}) / \text{Rnc} \\ &= (18.575 + 86.825) / 18.575 \\ &= 5.674 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (\text{Rn} + \text{tn}) \\ &= \pi/2 \cdot (18.575 + 86.825) \\ &= 165.562 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc}, \text{L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 86.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(\text{fy} \cdot \text{ky}, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot P \cdot \text{Rn}^2 \cdot \text{ky}^2) \\ &= \min(55347 \cdot 5.67, 1.5 \cdot 1306.0(192.738+0.0), \pi/4 \cdot 184.3 \cdot 18.58^2 \cdot 5.67^2) \\ &= 16084.039 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T})) \\ &= 16084.039 / (165.562(0.49 \cdot 7.071 + 0.6 \cdot 181.825 + 0.49 \cdot 0.0)) \\ &= 86.308 < \text{or} = \text{to } 1647.490 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-73 °C
User entered Minimum Design Temperature	:	-33 °C



The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : -20 °C
Temperature reduction per Graph 3.13 : 53 °C
The computed stress ratio is : 0.381

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (2.6 • 1.0)/(10.0 - 3.175 - 0.0)
= 0.381

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -33 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.030

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (2.6 • 1.0)/(90.0 - 3.175 - 0.0)
= 0.030

Gov. MDMT of the nozzle to shell joint welded assembly : -73 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 3.6491 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 488.6491 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: L32"

From: 30

Pressure for Reinforcement Calculations	P	184.342	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1647.49	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	3048.00	mm
Shell Finished (Minimum) Thickness	t	185.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.00	cm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		1.2126	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	184.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L32"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.213	in.
Actual Thickness Used in Calculation	0.394	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 37.15 \cdot (\exp(184.342/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0 \\ &= 2.8159 + 3.175 + 0.0 = 5.9909 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 6.825 + 37.15) / 37.15) - 0.0 \\ &= 408.69 - 0.0 = 408.69 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

$$\begin{aligned} \text{Min. required nozzle neck thickness, table 4.5.2 + c} &: 6.595 \text{ mm} \\ \text{Min. required shell thickness, internal p + c} &: 184.181 \text{ mm} \end{aligned}$$

$$\text{Nozzle neck minimum required thickness} : 6.595 \text{ mm}$$

Minimum nozzle neck thickness 10.008 mm, is greater than required thickness 6.595 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1306.0/1647.5, 1] \\ &= 0.793 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 90.0 - 3.175 \\ &= 86.825 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 10.0 - 3.175 \\ &= 6.825 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 3054.35/181.825 \\ &= 16.798 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3048.0/2 + 3.175 \\ &= 1527.175 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{1527.175 \cdot 181.825}, 2 \cdot 18.575) \\ &= 37.150 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\ &= \min[1.5 \cdot 181.825, 0.0] + \sqrt{18.575 \cdot 86.825} \\ &= 40.159 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.825 + 0.0) \\ &= 1454.600 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$



$$= \min[40.159, 300.0, 1454.6] + 181.825 \\ = 221.984 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR) \\ = 181.825 + ((0.0 \cdot 1.0) / 37.15) \\ = 181.825 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\ = \min(12, (2 \cdot 18.58 + 86.825) / (\sqrt{(3054.35 + 181.825) \cdot 181.825})) \\ = 0.162$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ = 181.825 \cdot 37.15 \cdot \max((0.162/5)^{0.85}, 1) \\ = 67.548 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH \\ = 86.825 \cdot 221.984 \\ = 192.738 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg41^2 \\ = 0.5 \cdot 10.0^2 \\ = 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ = 67.548 + 0.793(192.738 + 0.0) + 0.5 + 0.0 + 0.0 \\ = 220.835 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn] \\ = 86.825 / \ln[1 + 86.825/18.575] \\ = 50.016 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff] \\ = 181.825 / \ln[1 + 181.825/1527.175] \\ = 1616.384 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E \\ = 1.5 \cdot 1647.49 \cdot 1.0 \\ = 2471.2 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH \\ = 184.342 \cdot 50.016 \cdot 221.984 \\ = 20467.0 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn) \\ = 184.342 \cdot 1616.384 \cdot (37.15 + 86.825) \\ = 369404.9 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc \\ = 184.342 \cdot 1616.384 \cdot 18.575 \\ = 55347.4 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P \\ = (369404.9 + 55347.4 + 20467.0) / 184.342 \\ = 2415.2 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff}) \\ &= 2471.235 / (2 \cdot 2415.183/220.835 - 1616.384/181.825) \\ &= 190.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= \text{S}[t/\text{Rxs}] \\ &= 1647.49[181.825/1616.384] \\ &= 185.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(\text{Pmax1}, \text{Pmax2}) \\ &= \min(190.338, 185.324) \\ &= 185.324 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (\text{fN} + \text{fS} + \text{fY}) / \text{AT} \\ &= (20467.037 + 369404.875 + 55347.414) / 220.835 \\ &= 2016.072 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= \text{P} \cdot \text{Rxs} / \text{teff} \\ &= 184.342 \cdot 1616.384/181.825 \\ &= 1638.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 2016.072 - 1638.76, 1638.76) \\ &= 2393.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2471.24	kgf/cm ²
Local Primary Membrane Stress	PL	2393.38	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	185.32	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (\text{Rnc} + \text{tn}) / \text{Rnc} \\ &= (18.575 + 86.825) / 18.575 \\ &= 5.674 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (\text{Rn} + \text{tn}) \\ &= \pi/2 \cdot (18.575 + 86.825) \\ &= 165.562 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 86.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(\text{fy} \cdot \text{ky}, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot \text{P} \cdot \text{Rn}^2 \cdot \text{ky}^2) \\ &= \min(55347 \cdot 5.67, 1.5 \cdot 1306.0(192.738 + 0.0), \pi/4 \cdot 184.3 \cdot 18.58^2 \cdot 5.67^2) \\ &= 16084.039 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T})) \\ &= 16084.039 / (165.562(0.49 \cdot 7.071 + 0.6 \cdot 181.825 + 0.49 \cdot 0.0)) \\ &= 86.308 < \text{or} = \text{to } 1647.490 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-73 °C
User entered Minimum Design Temperature	:	-33 °C



The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : -20 °C
Temperature reduction per Graph 3.13 : 53 °C
The computed stress ratio is : 0.381

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (2.6 • 1.0)/(10.0 - 3.175 - 0.0)
= 0.381

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -33 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.030

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (2.6 • 1.0)/(90.0 - 3.175 - 0.0)
= 0.030

Gov. MDMT of the nozzle to shell joint welded assembly : -73 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 3.6491 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 488.6491 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: L52"

From: 30

Pressure for Reinforcement Calculations	P	184.342	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1647.49	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	3048.00	mm
Shell Finished (Minimum) Thickness	t	185.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		450.00	cm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		1.2126	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	184.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L52"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.213	in.
Actual Thickness Used in Calculation	0.394	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 37.15 \cdot (\exp(184.342/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0 \\ &= 2.8159 + 3.175 + 0.0 = 5.9909 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 6.825 + 37.15) / 37.15) - 0.0 \\ &= 408.69 - 0.0 = 408.69 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

$$\begin{aligned} \text{Min. required nozzle neck thickness, table 4.5.2 + c} &: 6.595 \text{ mm} \\ \text{Min. required shell thickness, internal p + c} &: 184.181 \text{ mm} \end{aligned}$$

$$\text{Nozzle neck minimum required thickness} : 6.595 \text{ mm}$$

Minimum nozzle neck thickness 10.008 mm, is greater than required thickness 6.595 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1306.0/1647.5, 1] \\ &= 0.793 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 90.0 - 3.175 \\ &= 86.825 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 10.0 - 3.175 \\ &= 6.825 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 3054.35/181.825 \\ &= 16.798 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3048.0/2 + 3.175 \\ &= 1527.175 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{(Reff \cdot t)}, 2 \cdot R_n) \\ &= \min(\sqrt{(1527.175 \cdot 181.825)}, 2 \cdot 18.575) \\ &= 37.150 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 181.825, 0.0] + \sqrt{(18.575 \cdot 86.825)} \\ &= 40.159 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.825 + 0.0) \\ &= 1454.600 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$



$$= \min[40.159, 300.0, 1454.6] + 181.825 \\ = 221.984 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR) \\ = 181.825 + ((0.0 \cdot 1.0) / 37.15) \\ = 181.825 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\ = \min(12, (2 \cdot 18.58 + 86.825) / (\sqrt{(3054.35 + 181.825) \cdot 181.825})) \\ = 0.162$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ = 181.825 \cdot 37.15 \cdot \max((0.162/5)^{0.85}, 1) \\ = 67.548 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH \\ = 86.825 \cdot 221.984 \\ = 192.738 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2 \\ = 0.5 \cdot 10.0^2 \\ = 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ = 67.548 + 0.793(192.738 + 0.0) + 0.5 + 0.0 + 0.0 \\ = 220.835 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn] \\ = 86.825 / \ln[1 + 86.825/18.575] \\ = 50.016 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff] \\ = 181.825 / \ln[1 + 181.825/1527.175] \\ = 1616.384 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E \\ = 1.5 \cdot 1647.49 \cdot 1.0 \\ = 2471.2 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH \\ = 184.342 \cdot 50.016 \cdot 221.984 \\ = 20467.0 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn) \\ = 184.342 \cdot 1616.384 \cdot (37.15 + 86.825) \\ = 369404.9 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc \\ = 184.342 \cdot 1616.384 \cdot 18.575 \\ = 55347.4 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P \\ = (369404.9 + 55347.4 + 20467.0) / 184.342 \\ = 2415.2 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff}) \\ &= 2471.235 / (2 \cdot 2415.183/220.835 - 1616.384/181.825) \\ &= 190.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= \text{S}[t/\text{Rxs}] \\ &= 1647.49[181.825/1616.384] \\ &= 185.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(\text{Pmax1}, \text{Pmax2}) \\ &= \min(190.338, 185.324) \\ &= 185.324 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (\text{fN} + \text{fS} + \text{fY}) / \text{AT} \\ &= (20467.037 + 369404.875 + 55347.414) / 220.835 \\ &= 2016.072 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot \text{Rxs} / \text{teff} \\ &= 184.342 \cdot 1616.384/181.825 \\ &= 1638.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 2016.072 - 1638.76, 1638.76) \\ &= 2393.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2471.24	kgf/cm ²
Local Primary Membrane Stress	PL	2393.38	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	185.32	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (\text{Rnc} + \text{tn}) / \text{Rnc} \\ &= (18.575 + 86.825) / 18.575 \\ &= 5.674 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (\text{Rn} + \text{tn}) \\ &= \pi/2 \cdot (18.575 + 86.825) \\ &= 165.562 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 86.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(\text{fy} \cdot \text{ky}, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot P \cdot \text{Rn}^2 \cdot \text{ky}^2) \\ &= \min(55347 \cdot 5.67, 1.5 \cdot 1306.0(192.738 + 0.0), \pi/4 \cdot 184.3 \cdot 18.58^2 \cdot 5.67^2) \\ &= 16084.039 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T})) \\ &= 16084.039 / (165.562(0.49 \cdot 7.071 + 0.6 \cdot 181.825 + 0.49 \cdot 0.0)) \\ &= 86.308 < \text{or} = \text{to } 1647.490 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-73 °C
User entered Minimum Design Temperature	:	-33 °C



The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : -20 °C
Temperature reduction per Graph 3.13 : 53 °C
The computed stress ratio is : 0.381

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (2.6 • 1.0)/(10.0 - 3.175 - 0.0)
= 0.381

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -33 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.030

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (2.6 • 1.0)/(90.0 - 3.175 - 0.0)
= 0.030

Gov. MDMT of the nozzle to shell joint welded assembly : -73 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 3.6491 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 488.6491 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: L72"

From: 30

Pressure for Reinforcement Calculations	P	184.342	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1647.49	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	3048.00	mm
Shell Finished (Minimum) Thickness	t	185.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		500.00	cm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		1.2126	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	184.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: L72"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.213	in.
Actual Thickness Used in Calculation	0.394	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 37.15 \cdot (\exp(184.342/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0 \\ &= 2.8159 + 3.175 + 0.0 = 5.9909 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 6.825 + 37.15) / 37.15) - 0.0 \\ &= 408.69 - 0.0 = 408.69 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

$$\begin{aligned} \text{Min. required nozzle neck thickness, table 4.5.2 + c} &: 6.595 \text{ mm} \\ \text{Min. required shell thickness, internal p + c} &: 184.181 \text{ mm} \end{aligned}$$

$$\text{Nozzle neck minimum required thickness} : 6.595 \text{ mm}$$

Minimum nozzle neck thickness 10.008 mm, is greater than required thickness 6.595 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1306.0/1647.5, 1] \\ &= 0.793 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 90.0 - 3.175 \\ &= 86.825 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 10.0 - 3.175 \\ &= 6.825 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 3054.35/181.825 \\ &= 16.798 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3048.0/2 + 3.175 \\ &= 1527.175 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{1527.175 \cdot 181.825}, 2 \cdot 18.575) \\ &= 37.150 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\ &= \min[1.5 \cdot 181.825, 0.0] + \sqrt{18.575 \cdot 86.825} \\ &= 40.159 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.825 + 0.0) \\ &= 1454.600 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$



$$= \min[40.159, 300.0, 1454.6] + 181.825 \\ = 221.984 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR) \\ = 181.825 + ((0.0 \cdot 1.0) / 37.15) \\ = 181.825 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\ = \min(12, (2 \cdot 18.58 + 86.825) / (\sqrt{(3054.35 + 181.825) \cdot 181.825})) \\ = 0.162$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ = 181.825 \cdot 37.15 \cdot \max((0.162/5)^{0.85}, 1) \\ = 67.548 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH \\ = 86.825 \cdot 221.984 \\ = 192.738 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2 \\ = 0.5 \cdot 10.0^2 \\ = 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ = 67.548 + 0.793(192.738 + 0.0) + 0.5 + 0.0 + 0.0 \\ = 220.835 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn] \\ = 86.825 / \ln[1 + 86.825/18.575] \\ = 50.016 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff] \\ = 181.825 / \ln[1 + 181.825/1527.175] \\ = 1616.384 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E \\ = 1.5 \cdot 1647.49 \cdot 1.0 \\ = 2471.2 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH \\ = 184.342 \cdot 50.016 \cdot 221.984 \\ = 20467.0 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn) \\ = 184.342 \cdot 1616.384 \cdot (37.15 + 86.825) \\ = 369404.9 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc \\ = 184.342 \cdot 1616.384 \cdot 18.575 \\ = 55347.4 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P \\ = (369404.9 + 55347.4 + 20467.0) / 184.342 \\ = 2415.2 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= \text{Sallow} / (2 \cdot \text{Ap} / \text{AT} - \text{Rxs} / \text{teff}) \\ &= 2471.235 / (2 \cdot 2415.183 / 220.835 - 1616.384 / 181.825) \\ &= 190.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= \text{S}[\text{t} / \text{Rxs}] \\ &= 1647.49[181.825 / 1616.384] \\ &= 185.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(\text{Pmax1}, \text{Pmax2}) \\ &= \min(190.338, 185.324) \\ &= 185.324 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (\text{fN} + \text{fS} + \text{fY}) / \text{AT} \\ &= (20467.037 + 369404.875 + 55347.414) / 220.835 \\ &= 2016.072 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= \text{P} \cdot \text{Rxs} / \text{teff} \\ &= 184.342 \cdot 1616.384 / 181.825 \\ &= 1638.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 2016.072 - 1638.76, 1638.76) \\ &= 2393.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2471.24	kgf/cm ²
Local Primary Membrane Stress	PL	2393.38	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	185.32	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (\text{Rnc} + \text{tn}) / \text{Rnc} \\ &= (18.575 + 86.825) / 18.575 \\ &= 5.674 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi / 2 \cdot (\text{Rn} + \text{tn}) \\ &= \pi / 2 \cdot (18.575 + 86.825) \\ &= 165.562 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 86.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(\text{fy} \cdot \text{ky}, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi / 4 \cdot \text{P} \cdot \text{Rn}^2 \cdot \text{ky}^2) \\ &= \min(55347 \cdot 5.67, 1.5 \cdot 1306.0(192.738 + 0.0), \pi / 4 \cdot 184.3 \cdot 18.58^2 \cdot 5.67^2) \\ &= 16084.039 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T})) \\ &= 16084.039 / (165.562(0.49 \cdot 7.071 + 0.6 \cdot 181.825 + 0.49 \cdot 0.0)) \\ &= 86.308 < \text{or} = \text{to } 1647.490 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-73 °C
User entered Minimum Design Temperature	:	-33 °C



The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : -20 °C
Temperature reduction per Graph 3.13 : 53 °C
The computed stress ratio is : 0.381

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$
$$= (2.6 \cdot 1.0) / (10.0 - 3.175 - 0.0)$$
$$= 0.381$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -33 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.030

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$
$$= (2.6 \cdot 1.0) / (90.0 - 3.175 - 0.0)$$
$$= 0.030$$

Gov. MDMT of the nozzle to shell joint welded assembly : -73 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 3.6491 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 488.6491 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: MH-124"

From: 60

Pressure for Reinforcement Calculations	P	184.342	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1647.49	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Hemispherical Head	D	3137.00	mm
Head Finished (Minimum) Thickness	t	94.0000	mm
Head Internal Corrosion Allowance	cas	3.1750	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		20.5354	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	44.0000	mm
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	156.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	94.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-124"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	20.535	in.
Actual Thickness Used in Calculation	1.732	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the



attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 527.95 \cdot (\exp(184.342/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0 \\ &= 40.018 + 3.175 + 0.0 = 43.193 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 40.825 + 527.95) / 527.95) - 0.0 \\ &= 187.805 - 0.0 = 187.8047 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1306.0/1647.5, 1] \\ &= 0.793 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 200.0 - 3.175 \\ &= 196.825 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 44.0 - 3.175 \\ &= 40.825 \text{ mm} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 3137.0/2 + 3.175 \\ &= 1571.675 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{1571.675 \cdot 90.825}, 2 \cdot 263.975) \\ &= 377.819 \text{ mm} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((90.825 + 0.0)/196.825)^{0.35}, 1) \\ &= \min(0.763, 1) \\ &= 0.763 \end{aligned}$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t) \\ &= \min(90.825 + 0.0 + 0.763 \cdot \sqrt{263.975 \cdot 196.825}, 500.0 + 90.825) \\ &= 264.711 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot frp) / LR) \\ &= 90.825 + ((0.0 \cdot 1.0) / 377.819) \\ &= 90.825 \text{ mm} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 90.825 \cdot 377.819 \\ &= 343.154 \text{ cm}^2 \end{aligned}$$



Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2B \\ &= 474.004 + 44.161 \\ &= 518.165 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 343.154 + 0.793(518.165 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 754.415 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 196.825 / \ln[1 + 196.825/263.975] \\ &= 353.297 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 90.825 / \ln[1 + 90.825/1571.675] \\ &= 1616.664 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1647.49 \cdot 1.0 \\ &= 2471.2 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 184.342 \cdot 353.297 \cdot 264.711 \\ &= 172399.5 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (L_R + t_n)/2 \\ &= 184.342 \cdot 1616.664 \cdot (377.819 + 196.825)/2 \\ &= 856274.2 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc})/2 \\ &= (184.342 \cdot 1616.664 \cdot 263.975)/2 \\ &= 393347.7 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N)/P \\ &= (856274 + 393347.7 + 172399.5)/184.342 \\ &= 7714.0 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p/AT - R_{xs}/(2 \cdot t_{eff})) \\ &= 2471.235 / (2 \cdot 7714.045/754.415 - 1616.664/(2 \cdot 90.825)) \\ &= 214.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S/[t/R_{xs}] \\ &= 2 \cdot 1647.49[90.825/1616.664] \\ &= 185.1 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(213.95, 185.114) \\ &= 185.114 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (172399.5 + 856274 + 393347.656)/754.415 \\ &= 1884.936 \text{ kgf/cm}^2 \end{aligned}$$



General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / (2 \cdot t_{eff}) \\ &= 184.342 \cdot 1616.664 / (2 \cdot 90.825) \\ &= 1640.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1884.936 - 1640.624, 1640.624) \\ &= 2129.2 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2471.24	kgf/cm ²
Local Primary Membrane Stress	PL	2129.25	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	185.11	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (263.975 + 196.825) / 263.975 \\ &= 1.746 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (263.975 + 196.825) \\ &= 723.823 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 196.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot ky, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot ky^2) \\ &= \min(393348 \cdot 1.75, 1.5 \cdot 1306.0(597.037 + 0.0), \pi/4 \cdot 184.3 \cdot 263.98^2 \cdot 1.75^2) \\ &= 307424.719 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 307424.719 / (723.823(0.49 \cdot 7.071 + 0.6 \cdot 90.825 + 0.49 \cdot 0.0)) \\ &= 732.791 < \text{ or } = \text{ to } 1647.490 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	2 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	7 °C
Temperature reduction per Graph 3.13	:	4 °C
The computed stress ratio is	:	0.905

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (36.953 \cdot 1.0) / (44.0 - 3.175 - 0.0) \\ &= 0.905 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	16 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	20 °C



Temperature reduction per Graph 3.13 : 4 °C
The computed stress ratio is : 0.917

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (83.316 • 1.0)/(94.0 - 3.175 - 0.0)
= 0.917

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 16 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 69.2150 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 663.2149 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: NozzleC10"

From: 70

Pressure for Reinforcement Calculations	P	184.342	kgf/cm ²
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1647.49	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Elliptical Head	D	508.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	30.0000	mm
Head Internal Corrosion Allowance	cas	3.1750	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 I	
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1306.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.4252	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: NozzleC10"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	8.425	in.
Actual Thickness Used in Calculation	0.787	in.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + ci + co$
 $= 0.5 \cdot 220.35 \cdot (\exp(184.342/(1306.0 \cdot 1.0)) - 1) + 3.175 + 0.0$
 $= 16.7023 + 3.175 + 0.0 = 19.8773 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:



$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P \cdot lq \\ &= 1306.0 \cdot 1.0 \cdot \ln((2 \cdot 16.825 + 220.35) / 220.35) - 0.0 \\ &= 185.605 - 0.0 = 185.6049 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.285 mm
Min. required shell thickness, internal p + c : 33.623 mm

Nozzle neck minimum required thickness : 11.285 mm

Minimum nozzle neck thickness 19.990 mm, is greater than required thickness 11.285 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 922.8,	Allowable	: 1306.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2468.4 kgf/cm ²	Passed
Occasional	: 560.7,	Allowable	: 1737.0 kgf/cm ²	Passed
Shear	: 235.7,	Allowable	: 914.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1306.0/1647.5, 1] \\ &= 0.793 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 80.0 - 3.175 \\ &= 76.825 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 20.0 - 3.175 \\ &= 16.825 \text{ mm} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2] \\ &= 0.9/6 \cdot 514.35 [2 + (514.35/(2 \cdot 130.175))^2] \\ &= 455.434 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{455.434 \cdot 26.825}, 2 \cdot 110.175) \\ &= 110.531 \text{ mm} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((26.825 + 0.0)/76.825)^{0.35}, 1) \\ &= \min(0.692, 1) \\ &= 0.692 \end{aligned}$$

Intermediate value [Cp]:

$$\begin{aligned} &= \exp((0.35 \cdot d_i - X_o) / (16t)) \\ &= \exp((0.35 \cdot 514.35 - 187.0) / (16 \cdot 26.825)) \\ &= 0.984 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t) \\ &= \min(26.825 + 0.0 + 0.692 \cdot \sqrt{110.175 \cdot 76.825}, 300.0 + 26.825) \\ &= 90.483 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot frp) / LR) \\ &= 26.825 + ((0.0 \cdot 1.0) / 110.531) \\ &= 26.825 \text{ mm} \end{aligned}$$



Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 26.825 \cdot 110.531 \\ &= 29.650 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2B \\ &= 59.021 + 9.56 \\ &= 68.581 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 29.65 + 0.793(68.581 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 84.515 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 76.825 / \ln[1 + 76.825/110.175] \\ &= 145.216 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 26.825 / \ln[1 + 26.825/455.434] \\ &= 468.718 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1647.495 \cdot 1.0 \\ &= 2471.2 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 184.342 \cdot 145.216 \cdot 90.483 \\ &= 24221.9 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (LR + t_n)/2 \\ &= 184.342 \cdot 468.718 \cdot (110.531 + 76.825)/2 \\ &= 80941.8 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc})/2 \\ &= (184.342 \cdot 468.718 \cdot 110.175)/2 \\ &= 47598.0 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (80941.8 + 47598.0 + 24221.9)/184.342 \\ &= 828.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff})) \\ &= 2471.242 / (2 \cdot 828.687 / 84.515 - 468.718 / (2 \cdot 26.825)) \\ &= 227.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S[t/R_{xs}] \\ &= 2 \cdot 1647.495[26.825/468.718] \\ &= 188.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$



$$= \min(227.267, 188.574)$$
$$= 188.574 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$
$$= (24221.9 + 80941.773 + 47598.043) / 84.515$$
$$= 1807.504 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$
$$= 184.342 \cdot 468.718 / (2 \cdot 26.825)$$
$$= 1610.5 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$
$$= \max(2 \cdot 1807.504 - 1610.523, 1610.523)$$
$$= 2004.5 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2471.24	kgf/cm ²
Local Primary Membrane Stress	PL	2004.49	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	188.57	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (110.175 + 76.825) / 110.175$$
$$= 1.697 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$
$$= \pi/2 \cdot (110.175 + 76.825)$$
$$= 293.739 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 76.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$
$$= \min(47598 \cdot 1.7, 1.5 \cdot 1306.0(82.468 + 0.0), \pi/4 \cdot 184.3 \cdot 110.17^2 \cdot 1.7^2)$$
$$= 50628.734 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$
$$= 50628.734 / (293.739(0.49 \cdot 7.071 + 0.6 \cdot 26.825 + 0.49 \cdot 0.0))$$
$$= 881.195 < \text{ or } = \text{ to } 1647.495 \text{ Weld Stress Passed}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle farther away from the intersecting weld.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated	
Computed Minimum Design Metal Temperature	: 13 °C
User entered Minimum Design Temperature	: -33 °C
The Curve for this Material is	: B
From Figure 3.7M- Non PWHT, the MDMT is	: 17 °C
Temperature reduction per Graph 3.12	: 4 °C
The computed stress ratio is	: 0.910

The computed thickness ratio [sr]:



$$= (tr \cdot E \cdot) / (tnom - ci - cext)$$

$$= (24.41 \cdot 1.0) / (30.0 - 3.175 - 0.0)$$

$$= 0.910$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 13 °C

The Drop for this Nozzle is : 37.8216 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 367.8216 mm

Input Echo, WRC107/537 Item 1, Description: NozzleC10" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.1750	mm
Vessel Diameter	Dv	914.400	mm
Vessel Thickness	Tv	30.000	mm
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm²
Vessel Hot S.I. Allowable	Smh	1647.49	kgf/cm²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.1750	mm
Nozzle Diameter	Dn	214.000	mm
Nozzle Thickness	Tn	80.000	mm
Nozzle Material		SA-765 I	
Nozzle UNS Number		K03046	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm²
Nozzle Hot S.I. Allowable	SNmh	1306.00	kgf/cm²
Design Internal Pressure	Dp	184.342	kgf/cm²
Include Pressure Thrust		No	

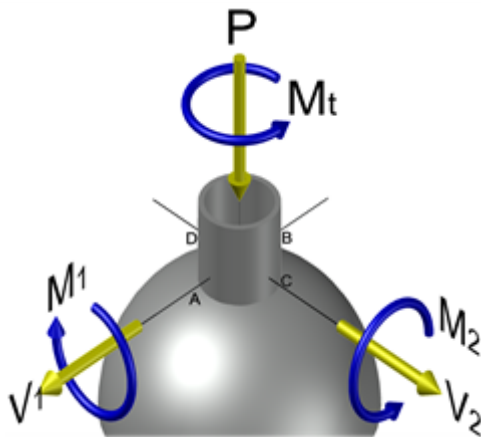
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-3185.5	kgf
Longitudinal Shear (SUS)	(V1) V1	2548.4	kgf
Circumferential Shear (SUS)	(Vc) V2	2548.4	kgf
Circumferential Moment (SUS)	(Mc) M1	1911.3	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	1911.3	kgf-m.
Torsional Moment (SUS)	Mt	2229.9	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 374.0 + 2 \cdot 1.65 \cdot \sqrt{(473.788 (30.0 - 3.175))}$$

$$= 746.028 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-3185.5	kgf
Circumferential Shear	(VC)	V2	2548.4	kgf
Longitudinal Shear	(VL)	V1	2548.4	kgf
Circumferential Moment	(MC)	M1	1911.3	kgf-m.
Longitudinal Moment	(ML)	M2	1911.3	kgf-m.
Torsional Moment		MT	2229.9	kgf-m.

Dimensionless Param: U = 1.66 TAU = 5.00 (1.93) RHO = 0.35

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.02287	(A,B,C,D)
$M(x) / P$	SP 1	0.02159	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.02150	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.02633	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.02150	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.02633	(A,B,C,D)
$N(y) * T / P$	SP 1	0.03231	(A,B,C,D)
$M(y) / P$	SP 1	0.00600	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.02734	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.00821	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.02734	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.00821	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	10.1	10.1	10.1	10.1	10.1	10.1	10.1	10.1
Rad. Bend.	P	57.4	-57.4	57.4	-57.4	57.4	-57.4	57.4	-57.4
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-50.6	-50.6	50.6	50.6
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-372.2	372.2	372.2	-372.2
Rad. Memb.	ML	-50.6	-50.6	50.6	50.6	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-372.2	372.2	372.2	-372.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-355.4	274.3	490.4	-368.8	-355.4	274.3	490.4	-368.8



Tang. Memb. P	14.3	14.3	14.3	14.3	14.3	14.3	14.3	14.3
Tang. Bend. P	15.9	-15.9	15.9	-15.9	15.9	-15.9	15.9	-15.9
Tang. Memb. MC	0.0	0.0	0.0	0.0	-64.4	-64.4	64.4	64.4
Tang. Bend. MC	0.0	0.0	0.0	0.0	-116.1	116.1	116.1	-116.1
Tang. Memb. ML	-64.4	-64.4	64.4	64.4	0.0	0.0	0.0	0.0
Tang. Bend. ML	-116.1	116.1	116.1	-116.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-150.2	50.0	210.7	-53.3	-150.2	50.0	210.7	-53.3
Shear VC	16.2	16.2	-16.2	-16.2	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-16.2	-16.2	16.2	16.2
Shear MT	37.8	37.8	37.8	37.8	37.8	37.8	37.8	37.8
Tot. Shear	54.0	54.0	21.7	21.7	21.7	21.7	54.0	54.0
Str. Int.	368.7	286.7	492.0	370.3	357.7	276.4	500.4	377.8

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	1627.5	1627.5	1627.5	1627.5	1627.5	1627.5	1627.5	1627.5
Rad.	Pl (SUS)	-40.5	-40.5	60.8	60.8	-40.5	-40.5	60.8	60.8
Rad.	Q (SUS)	-314.9	314.9	429.6	-429.6	-314.9	314.9	429.6	-429.6
Long.	Pm (SUS)	1627.5	1627.5	1627.5	1627.5	1627.5	1627.5	1627.5	1627.5
Long.	Pl (SUS)	-50.1	-50.1	78.7	78.7	-50.1	-50.1	78.7	78.7
Long.	Q (SUS)	-100.1	100.1	132.0	-132.0	-100.1	100.1	132.0	-132.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	16.2	16.2	-16.2	-16.2	-16.2	-16.2	16.2	16.2
Shear	Q (SUS)	37.8	37.8	37.8	37.8	37.8	37.8	37.8	37.8
Pm (SUS)		1627.5	1627.5	1627.5	1627.5	1627.5	1627.5	1627.5	1627.5
Pm+Pl (SUS)		1599.1	1599.1	1715.7	1715.7	1599.1	1599.1	1715.7	1715.7
Pm+Pl+Q (Total)		1490.6	1914.2	2119.5	1575.7	1479.5	1903.9	2127.9	1583.2

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1627.51	1647.49	Passed
Pm+Pl (SUS)	1715.74	2471.24	Passed
Pm+Pl+Q (TOTAL)	2127.93	4947.71	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.